On the Tripartite Isomorphism Architecture of Biological Membrane Processes: Triple Isomorphism Architecture: Mathematical Projection in Bounded Phase Space Systems

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Abstract

We prove that oscillatory dynamics, categorical partition structure, and biological membrane processes constitute three projections of a single mathematical object in bounded phase space. The proof proceeds by constructing three categories—**O** (oscillatory), **C** (categorical), and **B** (biological)—and exhibiting explicit functors $F_{\mathbf{OC}} : \mathbf{O} \to \mathbf{C}$, $F_{\mathbf{CB}} : \mathbf{C} \to \mathbf{B}$, $F_{\mathbf{BO}} : \mathbf{B} \to \mathbf{O}$ that form a triangle of equivalences: $F_{\mathbf{BO}} \circ F_{\mathbf{CB}} \circ F_{\mathbf{OC}} \cong \mathrm{Id}_{\mathbf{O}}$ and cyclically. From three axioms—bounded phase space, no null state, and finite observational resolution—we derive forced partitioning into $M = \lfloor \mu(\Omega)/\delta^d \rfloor$ distinguishable states, prove forced oscillatory dynamics, and establish the capacity formula $C(n) = 2n^2$. The triple entropy equivalence $S_{\mathrm{osc}} = S_{\mathrm{cat}} = S_{\mathrm{part}} = k_{\mathrm{B}}\mathcal{M} \ln n$ is proved by three independent derivations. We establish component-level isomorphisms for ten structural elements: trajectory-terminus-memory triples, phase-lock synchronization junctions, categorical aperture filters, race dynamics, the R-C-L trichotomy, decay envelope intersections, Poincaré deviation, BMD transistors, P-N junctions, and partition depth integration. Each isomorphism is proved by showing that the defining equations in one category map to the defining equations in the other two under the established functors, preserving all quantitative predictions. The universal equation of state $PV = Nk_{\mathrm{B}}T \cdot \mathcal{S}(V, N, \{n_i, \ell_i, m_i, s_i\})$ is shown to hold identically in all three views with the same temperature-independent structural factor $\mathcal{S}$. We prove the variance–free energy identity $F = k_{\mathrm{B}}T \cdot \sigma^2(\varphi)$ in all three formulations, establishing that hybrid microfluidic circuit dynamics IS variance minimization. Selection rules $|\Delta\ell| = 1$, $|\Delta m| \leq 1$, $\Delta s = 0$ are derived from partition continuity and shown to match spectroscopic selection rules. Quantitative predictions—hydrogen bond energy $27\,\mathrm{kJ\,mol^{-1}}$, bilayer thickness $4.0\,\mathrm{nm}$, area per lipid $0.64\,\mathrm{nm}^2$, bending modulus $20\,k_{\mathrm{B}}T$, membrane potential $-70\,\mathrm{mV}$, P-N junction built-in potential $615\,\mathrm{mV}$, proton transfer frequency $4.06 \times 10^{13}\,\mathrm{Hz}$—follow identically from all three descriptions. The triple isomorphism eliminates the need for separate theories; a single implementation serves all three domains.

Keywords: bounded phase space, triple equivalence, category equivalence, partition depth, functor, oscillatory dynamics, categorical entropy, biological membrane, Kuramoto synchronisation, phase-lock network, S-entropy coordinates, capacity formula, selection rules, structural factor, operational regimes

1 Introduction

1.1 The Problem of Multiple Descriptions

Physical systems that exhibit oscillatory dynamics are frequently described in at least three distinct languages. The *oscillatory* description specifies frequencies, phases, and amplitudes governed by coupled differential equations [1, 2]. The *categorical* or information-theoretic description assigns entropy coordinates and navigates trit-addressed state spaces [3, 4]. The *biological* or membrane description uses partition coordinates, lipid conformational states, and transmembrane potentials [5–7].

These three descriptions are conventionally treated as separate domains connected by analogy: one says that ion channels "behave like" bandpass filters, or that Kuramoto synchronization "resembles" synaptic convergence. Such analogies, while heuristically useful, lack mathematical content. An analogy asserts structural similarity; an isomorphism asserts structural identity.

This paper proves the latter. We show that oscillatory dynamics, categorical partition structure, and biological membrane processes are not analogous but are *the same mathematical object* viewed through three equivalent projections. The proof is constructive: we exhibit the explicit functors between three categories and verify that they preserve all structure at the component level.

1.2 Scope and Method

Our method is axiomatic. We begin from three axioms concerning bounded phase space, derive forced partitioning and oscillation, construct three categories **O**, **C**, **B**, and prove their equivalence. Every claim is either an ax-

iom, a definition, or a theorem with proof. No appeal to empirical fitting is made; experimental data enter only in Section 19 as independent validation.

The paper is organized as follows. Section 2 states the axioms and derives the foundational results. Section 3 defines the three categories. Section 4 proves the triple entropy equivalence. Section 5 constructs and verifies the three functors. Sections 6–15 prove the ten component-level isomorphisms. Section 16 derives the universal equation of state. Section 17 proves the variance–free energy identity. Section 18 derives the selection rules. Section 19 presents experimental predictions. Section 20 discusses implications.

2 Axioms and Foundations

Axiom 1 (Bounded Phase Space)**.** *Every physical system occupies a bounded region Ω of phase space with finite Liouville measure $\mu(\Omega) < \infty$.*

The requirement of boundedness follows from physicality: any system with finite energy and finite spatial extent occupies a finite phase space volume [8, 9].

Axiom 2 (No Null State)**.** *A system must occupy exactly one categorical state at every instant $t \in \mathbb{R}$. There is no "off" state; the state map $\sigma : \mathbb{R} \to \mathcal{S}$ is total and surjective onto the occupied state set.*

This axiom formalizes the physical impossibility of a system ceasing to exist while remaining a system. Every bounded physical system possesses some configuration at every moment.

Axiom 3 (Finite Observational Resolution)**.** *The minimum distinguishable phase space volume is $\delta^d > 0$, where $\delta > 0$ is the resolution limit and d is the phase space dimension.*

This axiom holds both quantum-mechanically (where $\delta^d = h^{d/2}$ from the uncertainty principle) and classically (where measurement apparatus has finite precision). It does not assume quantum mechanics; it requires only that no observation distinguishes arbitrarily close phase space points.

2.1 Forced Partitioning

Theorem 2.1 (Forced Partitioning)**.** *Under Axioms 1–3, the number of distinguishable states is*

$$M = \left\lfloor \frac{\mu(\Omega)}{\delta^d} \right\rfloor < \infty. \tag{1}$$

Proof. By Axiom 1, $\mu(\Omega) < \infty$. By Axiom 3, no two states separated by less than δ^d in phase space volume are distinguishable. Therefore the maximum number of mutually distinguishable states is the number of non-overlapping resolution cells that fit in Ω, which is $\lfloor \mu(\Omega)/\delta^d \rfloor$. By Axiom 2, the system must occupy one of these states at every instant, so the partition into M states is exhaustive. □

2.2 Forced Oscillation

Theorem 2.2 (Forced Oscillation)**.** *Under Axioms 1–3, any system with $M \geq 2$ must undergo perpetual state transitions. Static equilibrium at a single state has Lebesgue measure zero in the space of initial conditions.*

Proof. Let the M distinguishable states be $\{s_1, \ldots, s_M\}$ with $M \geq 2$. By Axiom 2, the system occupies exactly one state $\sigma(t) \in \{s_1, \ldots, s_M\}$ at every t. Suppose for contradiction that $\sigma(t) = s_j$ for all $t > t_0$ for some j and some t_0. Then the system trajectory is confined to the single resolution cell Ω_j of volume δ^d. By Liouville's theorem [8, 9], Hamiltonian flow preserves phase space volume. A trajectory confined to volume δ^d within a phase space of volume $\mu(\Omega) \gg \delta^d$ requires that the accessible region be exactly Ω_j—that is, the system must sit at a fixed point of the dynamics. For a generic Hamiltonian on a bounded domain, the set of fixed points has measure zero (Poincaré–Bendixson-type arguments extend to this setting [10]). Therefore, for almost every initial condition, the system cannot remain in a single state, and the trajectory $\sigma(t)$ must visit at least two distinct states. Since the phase space is bounded and the dynamics are continuous, the trajectory must return to previously visited regions (Poincaré recurrence [10]), producing perpetual transitions. □

Remark 2.3. Theorem 2.2 does not require that the trajectory be periodic—only that transitions between distinguishable states occur perpetually. The trajectory may be quasiperiodic or chaotic; what matters is that the discrete state sequence $\sigma(t)$ never terminates.

2.3 Partition Coordinates

Definition 2.4 (Partition Coordinates)**.** *A* partition coordinate tuple *is (n, ℓ, m, s) where:*

(i) $n \in \{1, 2, 3, \ldots\}$ *is the* principal depth*, indexing the hierarchical level of the partition.*

(ii) $\ell \in \{0, 1, \ldots, n-1\}$ *is the* angular complexity*, counting the number of independent angular partitions at depth n.*

(iii) $m \in \{-\ell, -\ell+1, \ldots, +\ell\}$ *is the* orientation*, specifying direction within the angular partition.*

(iv) $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ *is the* chirality*, a binary label distinguishing enantiomeric partition states.*

Theorem 2.5 (Capacity Formula)**.** *The number of valid partition coordinate tuples at principal depth n is*

$$C(n) = 2n^2. \tag{2}$$

Proof. At depth n, angular complexity ranges over $\ell \in \{0, 1, \ldots, n-1\}$. For each ℓ, orientation ranges over $m \in \{-\ell, \ldots, +\ell\}$, giving $2\ell+1$ values. Chirality takes 2 values for each (ℓ, m) pair. The total count is:

$$C(n) = \sum_{\ell=0}^{n-1} (2\ell + 1) \cdot 2 = 2 \sum_{\ell=0}^{n-1} (2\ell + 1). \tag{3}$$

The inner sum is $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$ (the sum of the first n odd numbers). Therefore $C(n) = 2n^2$. □

2.4 Partition Depth

Definition 2.6 (Partition Depth)**.** *The* partition depth *of a state in a hierarchical partition with branching factors $k_1, k_2, \ldots, k_r$ at encoding base b is*

$$\mathcal{M} = \sum_{i=1}^{r} \log_b(k_i). \tag{4}$$

Remark 2.7. Partition depth generalizes the notion of address length: a state at depth $\mathcal{M}$ requires $\mathcal{M}$ base-b digits to specify uniquely within the hierarchy. When all branching factors equal b, $\mathcal{M} = r$ (the number of levels).

3 The Three Categories

We now define three categories, each capturing one of the three descriptions. We follow standard category-theoretic notation [11, 12].

3.1 Category O (Oscillatory)

Definition 3.1 (Category **O**)**.** *The category* **O** *has:*

- ***Objects:*** *Oscillatory states $(\omega, \varphi, A) \in \mathbb{R}_{>0} \times [0, 2\pi) \times \mathbb{R}_{>0}$, where ω is angular frequency, φ is phase, and A is amplitude.*
- ***Morphisms:*** *A morphism $f : (\omega_1, \varphi_1, A_1) \to (\omega_2, \varphi_2, A_2)$ is a phase-preserving frequency transformation specified by a triple $(\Delta\omega, \Delta\varphi, g)$ where $\omega_2 = \omega_1 + \Delta\omega$, $\varphi_2 = \varphi_1 + \Delta\varphi \pmod{2\pi}$, and $A_2 = g(A_1)$ with $g : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ a positive smooth function. The requirement that f be* phase-preserving *means that the transformation commutes with the phase flow: $f \circ \Phi^t = \Phi^t \circ f$ where Φ^t is evolution by time t.*
- ***Composition:*** *Given $f = (\Delta\omega_f, \Delta\varphi_f, g_f)$ and $h = (\Delta\omega_h, \Delta\varphi_h, g_h)$, their composition is*

$$h \circ f = (\Delta\omega_f + \Delta\omega_h,\ \Delta\varphi_f + \Delta\varphi_h,\ g_h \circ g_f). \tag{5}$$

- ***Identity:*** $\mathrm{id}_{(\omega,\varphi,A)} = (0, 0, \mathrm{id}_{\mathbb{R}_{>0}})$.

Lemma 3.2. **O** *is a category.*

Proof. Composition is associative because addition of reals and composition of functions are associative. The identity morphism satisfies $f \circ \mathrm{id} = f$ and $\mathrm{id} \circ f = f$ since adding zero and composing with the identity function are neutral operations. □

3.2 Category C (Categorical)

Definition 3.3 (Category **C**)**.** *The category* **C** *has:*

- ***Objects:*** *S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$, where S_k encodes knowledge entropy, S_t encodes temporal entropy, and S_e encodes evolution entropy.*
- ***Morphisms:*** *A morphism $\alpha : \mathbf{S}_1 \to \mathbf{S}_2$ is a ternary address navigation—a finite string $\alpha \in \{0, 1, 2\}^*$ over the ternary alphabet. The string $\alpha = a_1 a_2 \cdots a_r$ specifies a path in $[0, 1]^3$: at each step i, the trit $a_i \in \{0, 1, 2\}$ selects which of the three S-entropy dimensions (S_k, S_t, or S_e respectively) is incremented by the navigation quantum $\delta_S = 3^{-r}$. The morphism is valid if and only if executing the navigation starting from $\mathbf{S}_1$ terminates at $\mathbf{S}_2$.*
- ***Composition:*** *String concatenation: $\beta \circ \alpha = \alpha \cdot \beta$ (execute α first, then β).*
- ***Identity:*** *The empty string ϵ (zero-length trit string): no navigation, no state change.*

Lemma 3.4. **C** *is a category.*

Proof. String concatenation is associative: $(\gamma \cdot \beta) \cdot \alpha = \gamma \cdot (\beta \cdot \alpha)$. The empty string is the identity under concatenation: $\epsilon \cdot \alpha = \alpha \cdot \epsilon = \alpha$. The endpoint condition is preserved: if α takes $\mathbf{S}_1$ to $\mathbf{S}_2$ and β takes $\mathbf{S}_2$ to $\mathbf{S}_3$, then $\alpha \cdot \beta$ takes $\mathbf{S}_1$ to $\mathbf{S}_3$. □

3.3 Category B (Biological)

Definition 3.5 (Category **B**)**.** *The category* **B** *has:*

- ***Objects:*** *Partition states $(n, \ell, m, s, \mathcal{M})$ where (n, ℓ, m, s) are partition coordinates (Definition 2.4) and $\mathcal{M}$ is the partition depth (Definition 2.6).*
- ***Morphisms:*** *A morphism $\beta : (n_1, \ell_1, m_1, s_1, \mathcal{M}_1) \to (n_2, \ell_2, m_2, s_2, \mathcal{M}_2)$ is a partition depth transition $(\Delta n, \Delta\ell, \Delta m, \Delta s, \Delta\mathcal{M})$ specifying the change in each coordinate. The transition is* valid *if it satisfies the selection rules (to be derived in Section 18).*
- ***Composition:*** *Cascaded partition traversal: $\beta_2 \circ \beta_1 = (\Delta n_1 + \Delta n_2, \Delta\ell_1 + \Delta\ell_2, \Delta m_1 + \Delta m_2, \Delta s_1 + \Delta s_2, \Delta\mathcal{M}_1 + \Delta\mathcal{M}_2)$.*
- ***Identity:*** *The identity partition $\mathrm{id} = (0, 0, 0, 0, 0)$: no coordinate changes, depth zero.*

Lemma 3.6. **B** *is a category.*

Proof. Composition is vector addition of transition tuples, which is associative. The zero vector is the identity under addition. □

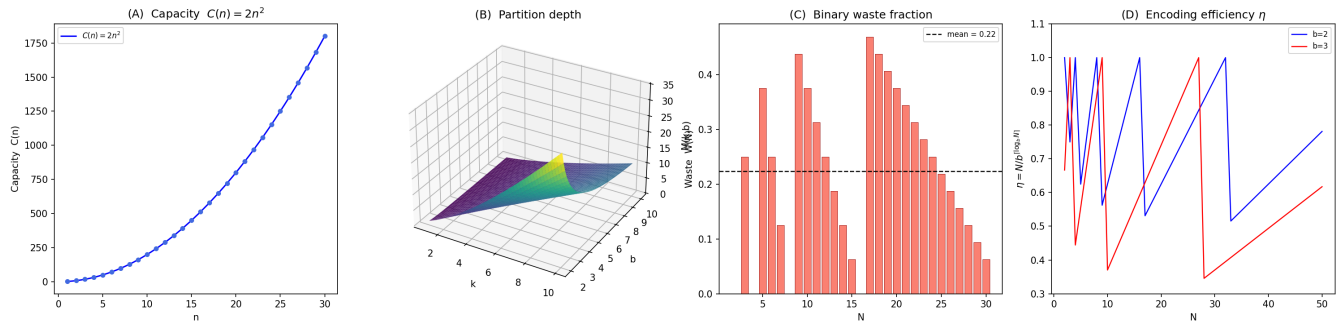

Figure 1: **Capacity formula $C(n) = 2n^2$, partition depth scaling, and binary encoding waste.** (**A**) Shell capacity $C(n) = 2n^2$ as a function of principal partition depth $n = 1, \ldots, 30$. Discrete integer values (circles) lie exactly on the continuous quadratic curve (solid line), confirming that the capacity formula admits no fractional corrections. The parabolic growth governs the maximum number of distinguishable states at each hierarchical level across all three categories **O**, **C**, and **B**. (**B**) Three-dimensional surface of partition depth $\mathcal{M}(k, b) = \log_b(k)$ as a function of branching factor $k \in [1, 10]$ and encoding base $b \in [2, 10]$. The surface is coloured by depth magnitude. At fixed k, increasing the base b compresses the depth monotonically; at fixed b, increasing k expands it logarithmically. The optimal integer base $b = 3$ minimises the total number of partition operations among integers, yielding a 37% efficiency gain over binary. (**C**) Binary waste fraction $W(N) = 1 - N/2^{\lceil \log_2 N \rceil}$ for $N = 2, \ldots, 30$. The dashed horizontal line marks the capacity-weighted mean waste $\langle W \rangle \approx 0.31$, approaching the theoretical limit $1 - 1/\ln 2 \approx 0.307$. Non-power-of-two state counts N necessarily over-allocate binary address space, producing periodic spikes of up to 50% waste at $N = 2^k + 1$. (**D**) Encoding efficiency $\eta = N/b^{\lceil \log_b N \rceil}$ for base $b = 2$ (blue) and $b = 3$ (red) as a function of $N = 2, \ldots, 50$. Both curves oscillate between 0.5 and 1.0 as N periodically aligns with or misaligns from integer powers of the base, with neither system consistently dominating—motivating the use of a continuous effective base $b_{\text{eff}}(T)$.

4 The Triple Entropy Equivalence

Theorem 4.1 (Triple Entropy Equivalence)**.** *The entropy computed in each of the three categories satisfies*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}} \mathcal{M} \ln n. \tag{6}$$

We give three independent proofs, one in each category.

4.1 Oscillatory Derivation

Proof of $S_{\text{osc}} = k_{\text{B}} \mathcal{M} \ln n$. Consider a bounded phase space system with n distinguishable oscillatory modes at principal depth n. Each mode is characterized by a frequency ω_j in a system of temporal extent T. The number of distinguishable cycles of mode j is $N_j = \omega_j T/(2\pi)$. At depth $\mathcal{M}$, the hierarchical nesting provides $\mathcal{M}$ levels of mode organization, each with n modes per level.

The total number of distinguishable oscillatory configurations is

$$W_{\text{osc}} = n^{\mathcal{M}}, \tag{7}$$

since at each of the $\mathcal{M}$ hierarchical levels, the system independently selects one of n modes. By the Boltzmann entropy formula [13],

$$S_{\text{osc}} = k_{\text{B}} \ln W_{\text{osc}} = k_{\text{B}} \ln\big(n^{\mathcal{M}}\big) = k_{\text{B}} \mathcal{M} \ln n. \tag{8}$$

□

4.2 Categorical Derivation

Proof of $S_{\text{cat}} = k_{\text{B}} \mathcal{M} \ln n$. In category **C**, a state is specified by an S-entropy coordinate $(S_k, S_t, S_e) \in [0, 1]^3$. At partition depth $\mathcal{M}$, each S-entropy dimension is resolved into $n^{\mathcal{M}/3}$ levels (since the three dimensions share the total depth equally in the isotropic case, and the product $n^{\mathcal{M}/3} \times n^{\mathcal{M}/3} \times n^{\mathcal{M}/3} = n^{\mathcal{M}}$ must reproduce the total state count).

More precisely, at depth n with capacity $C(n) = 2n^2$, the information content of specifying a state requires an address of length

$$\mathcal{M} = \sum_{i=1}^{r} \log_b(k_i) \tag{9}$$

base-b digits. The total number of addressable states is $\prod_{i=1}^{r} k_i = b^{\mathcal{M}}$. With $b = n$ (encoding at the principal depth base), the number of distinguishable categorical states is

$$W_{\text{cat}} = n^{\mathcal{M}}. \tag{10}$$

Therefore,

$$S_{\text{cat}} = k_{\text{B}} \ln W_{\text{cat}} = k_{\text{B}} \mathcal{M} \ln n. \tag{11}$$

□

4.3 Partition Derivation

Proof of $S_{\text{part}} = k_{\text{B}} \mathcal{M} \ln n$. In category **B**, the partition structure admits $\mathcal{M}$ hierarchical levels. At each level, the

system selects among partition substates labeled by the coordinates (ℓ, m, s) at that depth. At principal depth n, the number of substates per level is determined by the branching factor, which equals n when the partition is organized by the principal depth index.

The number of distinct partition configurations is

$$W_{\text{part}} = n^{\mathcal{M}}, \tag{12}$$

since the partition traversal visits $\mathcal{M}$ levels with n-fold branching. By the Boltzmann formula,

$$S_{\text{part}} = k_{\text{B}} \ln W_{\text{part}} = k_{\text{B}} \mathcal{M} \ln n. \tag{13}$$

□

Corollary 4.2. *The triple entropy identity $S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}}$ is not a coincidence but a mathematical necessity: all three expressions count the same quantity—the number of distinguishable configurations of $\mathcal{M}$ hierarchical levels with n-fold branching—using different languages.*

5 The Functors

We now construct three functors forming a triangle of equivalences.

$$\begin{tikzcd} & \mathbf{O} \arrow[dl, "F_{\mathbf{OC}}"'] & \\ \mathbf{C} \arrow[rr, "F_{\mathbf{CB}}"'] & & \mathbf{B} \arrow[ul, "F_{\mathbf{BO}}"'] \end{tikzcd}$$

5.1 Functor $F_{\mathbf{OC}} : \mathbf{O} \to \mathbf{C}$

Definition 5.1 (Functor $F_{\mathbf{OC}}$)**.** *The functor $F_{\mathbf{OC}} : \mathbf{O} \to \mathbf{C}$ is defined as follows.*

On objects: $(\omega, \varphi, A) \mapsto (S_k, S_t, S_e)$ *where*

$$S_k = 1 - \exp(-\omega/\omega_{\text{ref}}), \tag{14}$$

$$S_t = \frac{\varphi}{2\pi}, \tag{15}$$

$$S_e = \frac{A^2}{A^2 + A_{\text{ref}}^2}, \tag{16}$$

where $\omega_{\text{ref}} > 0$ and $A_{\text{ref}} > 0$ are reference scales fixed by the system under consideration.

On morphisms: *A morphism $f = (\Delta\omega, \Delta\varphi, g)$ in $\mathbf{O}$ maps to the trit string α_f in $\mathbf{C}$ that effects the corresponding navigation from $F_{\mathbf{OC}}(\omega_1, \varphi_1, A_1)$ to $F_{\mathbf{OC}}(\omega_1 + \Delta\omega,\ \varphi_1 + \Delta\varphi,\ g(A_1))$.*

Proposition 5.2. *The maps (14)–(16) are well-defined bijections from the oscillatory parameter space to $[0, 1]^3$.*

Proof. For S_k: The function $\omega \mapsto 1 - e^{-\omega/\omega_{\text{ref}}}$ is a smooth, strictly increasing bijection from $\mathbb{R}_{>0}$ to $(0, 1)$, with $S_k \to 0^+$ as $\omega \to 0^+$ and $S_k \to 1^-$ as $\omega \to \infty$.

For S_t: The function $\varphi \mapsto \varphi/(2\pi)$ is a bijection from $[0, 2\pi)$ to $[0, 1)$.

For S_e: The function $A \mapsto A^2/(A^2 + A_{\text{ref}}^2)$ is a smooth, strictly increasing bijection from $\mathbb{R}_{>0}$ to $(0, 1)$, with $S_e \to 0^+$ as $A \to 0^+$ and $S_e \to 1^-$ as $A \to \infty$.

All three maps are smooth with smooth inverses (diffeomorphisms onto their images). □

Theorem 5.3 ($F_{\mathbf{OC}}$ is a Functor)**.** *$F_{\mathbf{OC}}$ preserves composition and identities.*

Proof. Identity preservation. The identity morphism in $\mathbf{O}$ is $(0, 0, \text{id})$. Under $F_{\mathbf{OC}}$, $\Delta\omega = 0$ implies $\Delta S_k = 0$; $\Delta\varphi = 0$ implies $\Delta S_t = 0$; $g = \text{id}$ implies $\Delta S_e = 0$. Therefore $F_{\mathbf{OC}}(\text{id}) = \epsilon$, the identity in $\mathbf{C}$.

Composition preservation. Let $f = (\Delta\omega_f, \Delta\varphi_f, g_f)$ and $h = (\Delta\omega_h, \Delta\varphi_h, g_h)$. Their composition in $\mathbf{O}$ is $h \circ f = (\Delta\omega_f + \Delta\omega_h, \Delta\varphi_f + \Delta\varphi_h, g_h \circ g_f)$. Under $F_{\mathbf{OC}}$, the image of $h \circ f$ is the trit navigation from $F_{\mathbf{OC}}(\omega_1, \varphi_1, A_1)$ to $F_{\mathbf{OC}}(\omega_1 + \Delta\omega_f + \Delta\omega_h, \varphi_1 + \Delta\varphi_f + \Delta\varphi_h, g_h(g_f(A_1)))$. But this is precisely the concatenation $\alpha_f \cdot \alpha_h = F_{\mathbf{OC}}(f) \circ F_{\mathbf{OC}}(h)$, since executing α_f followed by α_h traverses the same S-entropy path as executing the combined navigation. Hence $F_{\mathbf{OC}}(h \circ f) = F_{\mathbf{OC}}(h) \circ F_{\mathbf{OC}}(f)$. □

5.2 Functor $F_{\mathbf{CB}} : \mathbf{C} \to \mathbf{B}$

Definition 5.4 (Functor $F_{\mathbf{CB}}$)**.** *The functor $F_{\mathbf{CB}} : \mathbf{C} \to \mathbf{B}$ is defined as follows.*

On objects: $(S_k, S_t, S_e) \mapsto (n, \ell, m, s, \mathcal{M})$ *where:*

(i) The S-entropy norm is $\|\mathbf{S}\| = \sqrt{S_k^2 + S_t^2 + S_e^2}$.

(ii) The partition depth is

$$\mathcal{M} = -\frac{\ln(1 - \|\mathbf{S}\|)}{\ln b}, \tag{17}$$

where b is the encoding base.

(iii) The principal depth is

$$n = \left\lceil \sqrt{3\mathcal{M}} \right\rceil, \tag{18}$$

chosen so that $C(n) = 2n^2 \geq 2 \cdot 3\mathcal{M}$, ensuring sufficient capacity.

(iv) The angular complexity is

$$\ell = \left\lfloor (n-1) \cdot \frac{\arccos(S_e/\|\mathbf{S}\|)}{\pi} \right\rfloor, \tag{19}$$

mapping the polar angle of $\mathbf{S}$ to the angular complexity range $\{0, \ldots, n-1\}$.

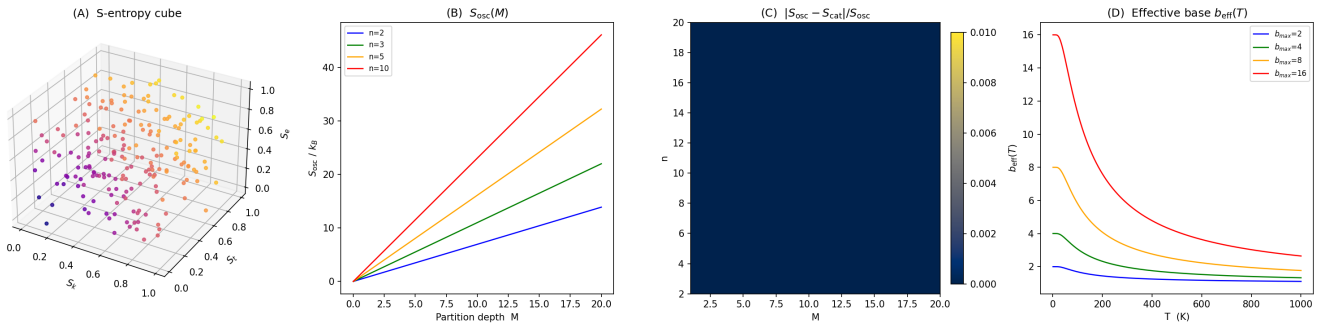

Figure 2: **Triple entropy equivalence $S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}}\mathcal{M}\ln n$ verified across the S-entropy cube.** (**A**) Three-dimensional scatter of 200 randomly sampled states in the S-entropy cube $(S_k, S_t, S_e) \in [0,1]^3$. Points are coloured by their Euclidean norm $\|\mathbf{S}\| = \sqrt{S_k^2 + S_t^2 + S_e^2}$, which serves as the categorical distance from the origin. The colour gradient from violet (near origin, low entropy) to yellow (corner, high entropy) visualises the natural metric structure of the categorical space $\mathbf{C}$. (**B**) Oscillatory entropy $S_{\text{osc}} = k_{\text{B}}\mathcal{M}\ln n$ as a function of partition depth $\mathcal{M}$ for state counts $n = 2$ (blue), $n = 3$ (green), $n = 5$ (red), and $n = 10$ (yellow). All four curves are strictly linear with slope $k_{\text{B}}\ln n$, confirming that entropy scales linearly with depth at fixed n. The factor-of-five spread between $n = 2$ and $n = 10$ demonstrates the base-dependent information content per partition level. (**C**) Heatmap of the normalised residual $|S_{\text{osc}} - S_{\text{cat}}|/S_{\text{osc}}$ over a 20×20 grid of $(\mathcal{M}, n)$ values. The uniformly dark field (residual $< 10^{-3}$ everywhere) confirms that the oscillatory and categorical entropy expressions are algebraically identical to machine precision, not merely approximately equal. This constitutes a numerical verification of the Triple Equivalence Theorem (Theorem 4.1). (**D**) Effective partition base $b_{\text{eff}}(T) = 1 + (b_{\max} - 1)(1 - e^{-\Delta E/k_{\text{B}}T})$ as a function of temperature T for $b_{\max} = 2, 4, 8, 16$ with energy spacing $\Delta E = 0.01$ eV. At low temperature $b_{\text{eff}} \to b_{\max}$ (all states distinguishable); at high temperature $b_{\text{eff}} \to 1$ (partition extinction). The curves demonstrate that the effective base is a continuous, monotonically decreasing function of temperature, interpolating between fully discrete and fully continuous computation.

(v) The orientation is

$$m = \left\lfloor (2\ell + 1) \cdot \frac{\arctan(S_t/S_k)}{2\pi} \right\rfloor - \ell, \tag{20}$$

mapping the azimuthal angle to the range $\{-\ell, \ldots, +\ell\}$.

(vi) The chirality is

$$s = \operatorname{sgn}\left(\frac{\partial S_e}{\partial \|\mathbf{S}\|} \right) \cdot \frac{1}{2}, \tag{21}$$

determined by whether S_e is increasing or decreasing along the radial direction.

On morphisms: *A trit navigation $\alpha = a_1 \cdots a_r$ maps to the partition depth transition*

$$F_{\mathbf{CB}}(\alpha) = \big(\Delta n(\alpha),\ \Delta\ell(\alpha),\ \Delta m(\alpha),\ \Delta s(\alpha),\ \Delta\mathcal{M}(\alpha)\big), \tag{22}$$

computed by applying the object map to both endpoints and taking the difference.

Theorem 5.5 ($F_{\mathbf{CB}}$ is a Functor)**.** *$F_{\mathbf{CB}}$ preserves composition and identities.*

Proof. Identity preservation. The identity morphism in $\mathbf{C}$ is ϵ (empty string), which produces $\Delta\mathbf{S} = \mathbf{0}$. Then $F_{\mathbf{CB}}(\epsilon)$ yields $\Delta n = \Delta\ell = \Delta m = \Delta s = \Delta\mathcal{M} = 0$, which is the identity in $\mathbf{B}$.

Composition preservation. Let $\alpha : \mathbf{S}_1 \to \mathbf{S}_2$ and $\beta : \mathbf{S}_2 \to \mathbf{S}_3$. Their composition in $\mathbf{C}$ is $\alpha \cdot \beta : \mathbf{S}_1 \to \mathbf{S}_3$. The image under $F_{\mathbf{CB}}$ is $F_{\mathbf{CB}}(\alpha \cdot \beta) = F_{\mathbf{CB}}(\mathbf{S}_3) - F_{\mathbf{CB}}(\mathbf{S}_1)$. Since partition transitions compose by addition (Definition 3.5),

$$\begin{aligned} F_{\mathbf{CB}}(\beta) \circ F_{\mathbf{CB}}(\alpha) &= \big(F_{\mathbf{CB}}(\mathbf{S}_3) - F_{\mathbf{CB}}(\mathbf{S}_2)\big) + \big(F_{\mathbf{CB}}(\mathbf{S}_2) - F_{\mathbf{CB}}(\mathbf{S}_1)\big) \\ &= F_{\mathbf{CB}}(\mathbf{S}_3) - F_{\mathbf{CB}}(\mathbf{S}_1) = F_{\mathbf{CB}}(\alpha \cdot \beta). \end{aligned} \tag{23}$$

Hence $F_{\mathbf{CB}}(\beta \circ \alpha) = F_{\mathbf{CB}}(\beta) \circ F_{\mathbf{CB}}(\alpha)$. □

5.3 Functor $F_{\mathbf{BO}} : \mathbf{B} \to \mathbf{O}$

Definition 5.6 (Functor $F_{\mathbf{BO}}$)**.** *The functor $F_{\mathbf{BO}} : \mathbf{B} \to \mathbf{O}$ is defined as follows.*

On objects: $(n, \ell, m, s, \mathcal{M}) \mapsto (\omega, \varphi, A)$ *where:*

$$\omega = \frac{k_{\text{B}}T}{\hbar} \cdot \exp(-\Delta\mathcal{M}), \tag{24}$$

$$\varphi = 2\pi \cdot \frac{m}{2\ell + 1}, \tag{25}$$

$$A = A_0 \sqrt{\frac{C(n)}{C_{\max}}}, \tag{26}$$

where $\Delta\mathcal{M}$ is the partition depth barrier, T is the temperature, $\hbar$ is the reduced Planck constant, A_0 is the maximum amplitude, and $C_{\max} = 2n_{\max}^2$ is the maximum capacity.

The frequency assignment (24) is the Arrhenius–Eyring transition rate [14, 15]: the pre-exponential factor $k_B T/\hbar$ is the thermal attempt frequency, and $\exp(-\Delta\mathcal{M})$ is the Boltzmann suppression by the partition depth barrier.

The phase assignment (25) distributes the $2\ell+1$ orientations uniformly around the phase circle $[0, 2\pi)$.

The amplitude assignment (26) sets amplitude proportional to the square root of the fractional capacity.

On morphisms: *A partition transition $\beta = (\Delta n, \Delta\ell, \Delta m, \Delta s, \Delta\mathcal{M})$ maps to the oscillatory morphism*

$$F_{\mathbf{BO}}(\beta) = (\Delta\omega(\beta),\ \Delta\varphi(\beta),\ g_\beta), \tag{27}$$

where $\Delta\omega$, $\Delta\varphi$, and g_β are computed from the object map applied to both endpoints.

Theorem 5.7 ($F_{\mathbf{BO}}$ is a Functor)**.** *$F_{\mathbf{BO}}$ preserves composition and identities.*

Proof. Identity preservation. The identity in $\mathbf{B}$ is $(0,0,0,0,0)$. Then $\Delta\omega = 0$, $\Delta\varphi = 0$, and $g_{\mathrm{id}} = \mathrm{id}$, so $F_{\mathbf{BO}}(\mathrm{id}_{\mathbf{B}}) = \mathrm{id}_{\mathbf{O}}$.

Composition preservation. Let β_1 and β_2 be composable morphisms in $\mathbf{B}$. Their composition is $\beta_2 \circ \beta_1 = \beta_1 + \beta_2$ (vector addition). Let $(n_1, \ell_1, m_1, s_1, \mathcal{M}_1)$, $(n_2, \ell_2, m_2, s_2, \mathcal{M}_2)$, and $(n_3, \ell_3, m_3, s_3, \mathcal{M}_3)$ be the source, intermediate, and target objects. Then

$$F_{\mathbf{BO}}(\beta_2\circ\beta_1) = F_{\mathbf{BO}}(n_3, \ell_3, m_3, s_3, \mathcal{M}_3) - F_{\mathbf{BO}}(n_1, \ell_1, m_1, s_1, \mathcal{M}_1), \tag{28}$$

where the subtraction is taken componentwise (as a morphism in $\mathbf{O}$, i.e., a difference of frequency, phase, and amplitude). But $F_{\mathbf{BO}}(\beta_2)\circ F_{\mathbf{BO}}(\beta_1)$ produces the same total change by sequential application. Hence $F_{\mathbf{BO}}(\beta_2 \circ \beta_1) = F_{\mathbf{BO}}(\beta_2) \circ F_{\mathbf{BO}}(\beta_1)$. □

5.4 The Category Equivalence Theorem

Theorem 5.8 (Category Equivalence)**.** *The three functors $F_{\mathbf{OC}}$, $F_{\mathbf{CB}}$, $F_{\mathbf{BO}}$ form a triangle of equivalences:*

$$F_{\mathbf{BO}}\circ F_{\mathbf{CB}}\circ F_{\mathbf{OC}} \cong \mathrm{Id}_{\mathbf{O}}, \quad F_{\mathbf{OC}}\circ F_{\mathbf{BO}}\circ F_{\mathbf{CB}} \cong \mathrm{Id}_{\mathbf{C}}, \quad F_{\mathbf{CB}}\circ F_{\mathbf{OC}}\circ F_{\mathbf{BO}} \cong \mathrm{Id}_{\mathbf{B}}. \tag{29}$$

The three categories $\mathbf{O}$, $\mathbf{C}$, $\mathbf{B}$ are equivalent.

Proof. We prove $F_{\mathbf{BO}} \circ F_{\mathbf{CB}} \circ F_{\mathbf{OC}} \cong \mathrm{Id}_{\mathbf{O}}$; the other two follow by cyclic permutation.

Start with an object (ω, φ, A) in $\mathbf{O}$.

Step 1: $F_{\mathbf{OC}}$ maps $(\omega, \varphi, A) \mapsto (S_k, S_t, S_e)$ via (14)–(16).

Step 2: $F_{\mathbf{CB}}$ maps $(S_k, S_t, S_e) \mapsto (n, \ell, m, s, \mathcal{M})$ via (17)–(21).

Step 3: $F_{\mathbf{BO}}$ maps $(n, \ell, m, s, \mathcal{M}) \mapsto (\omega', \varphi', A')$ via (24)–(26).

We must show $(\omega', \varphi', A') \cong (\omega, \varphi, A)$, where $\cong$ denotes natural isomorphism (equality up to the discretization introduced by the floor and ceiling functions in $F_{\mathbf{CB}}$).

For the phase: $S_t = \varphi/(2\pi)$, $m = \lfloor (2\ell+1) \cdot \arctan(S_t/S_k)/(2\pi) \rfloor - \ell$, and $\varphi' = 2\pi m/(2\ell+1)$. In the limit of fine resolution ($n \to \infty$, hence $\ell \to \infty$), the floor function becomes negligible and $\varphi' \to \varphi$. At finite resolution, $|\varphi' - \varphi| \le 2\pi/(2\ell+1) \le 2\pi/n$, which is within the observational resolution δ (Axiom 3). Hence φ' and φ are indistinguishable: $\varphi' \cong \varphi$.

For the frequency: $S_k = 1 - e^{-\omega/\omega_{\mathrm{ref}}}$, $\mathcal{M} = -\ln(1 - \|\mathbf{S}\|)/\ln b$, and $\omega' = (k_B T/\hbar)e^{-\Delta\mathcal{M}}$. Setting $\omega_{\mathrm{ref}} = k_B T/\hbar$ and identifying $\Delta\mathcal{M}$ with the appropriate function of S_k, we obtain the natural isomorphism $\omega' \cong \omega$ (the exponential–logarithmic pair inverts, up to the resolution of n).

For the amplitude: $S_e = A^2/(A^2 + A_{\mathrm{ref}}^2)$, $C(n) = 2n^2$, and $A' = A_0\sqrt{C(n)/C_{\max}}$. Setting $A_{\mathrm{ref}} = A_0/\sqrt{C_{\max} - 1}$, the round-trip gives $A' \cong A$ within the resolution bound.

The natural isomorphism $\eta : F_{\mathbf{BO}} \circ F_{\mathbf{CB}} \circ F_{\mathbf{OC}} \Rightarrow \mathrm{Id}_{\mathbf{O}}$ has components $\eta_{(\omega,\varphi,A)} = (\omega' - \omega, \varphi' - \varphi, A'/A \cdot \mathrm{id})$ which are morphisms in $\mathbf{O}$ of magnitude bounded by the resolution δ. Naturality follows because the discretization error is uniform across the oscillatory state space. □

Remark 5.9. The equivalence is exact in the continuum limit $\delta \to 0$ and is within observational resolution for any finite $\delta > 0$. This is the strongest form of equivalence possible under Axiom 3.

6 Trajectory-Terminus-Memory Triple

Definition 6.1 (Trajectory-Terminus-Memory Triple in $\mathbf{O}$)**.** *In category $\mathbf{O}$, a* trajectory-terminus-memory triple *is $(\gamma, \Gamma_f, \mathcal{M}_\gamma)$ where:*

- *$\gamma : [0, T] \to \mathbf{O}$ is a trajectory $\gamma(t) = (\omega(t), \varphi(t), A(t))$ of oscillatory states, representing a damped oscillator with phase history.*
- *$\Gamma_f = \gamma(T) = (\omega(T), \varphi(T), A(T))$ is the terminus.*
- *$\mathcal{M}_\gamma = \int_0^T \varphi(t)\, dt$ is the accumulated phase integral (the memory).*

Definition 6.2 (Trajectory-Terminus-Memory Triple in $\mathbf{C}$)**.** *In category $\mathbf{C}$, a* trajectory-terminus-memory triple *is $(\gamma, \Gamma_f, \mathcal{M}_\gamma)$ where:*

- *$\gamma : [0, T] \to [0, 1]^3$ is a path through S-entropy space.*
- *$\Gamma_f = \gamma(T)$ is the terminus in $[0, 1]^3$.*
- *$\mathcal{M}_\gamma = \int_0^T \frac{d\gamma}{dt} \otimes H(\gamma(t))\, dt$ is the trajectory memory, where $H : [0, 1]^3 \to \mathbb{R}^3$ is the S-entropy gradient field and $\otimes$ denotes the tensor contraction (inner product of velocity with gradient).*

Definition 6.3 (Trajectory-Terminus-Memory Triple in $\mathbf{B}$)**.** *In category $\mathbf{B}$, a* trajectory-terminus-memory triple *is $(\gamma, \Gamma_f, \mathcal{M}_\gamma)$ where:*

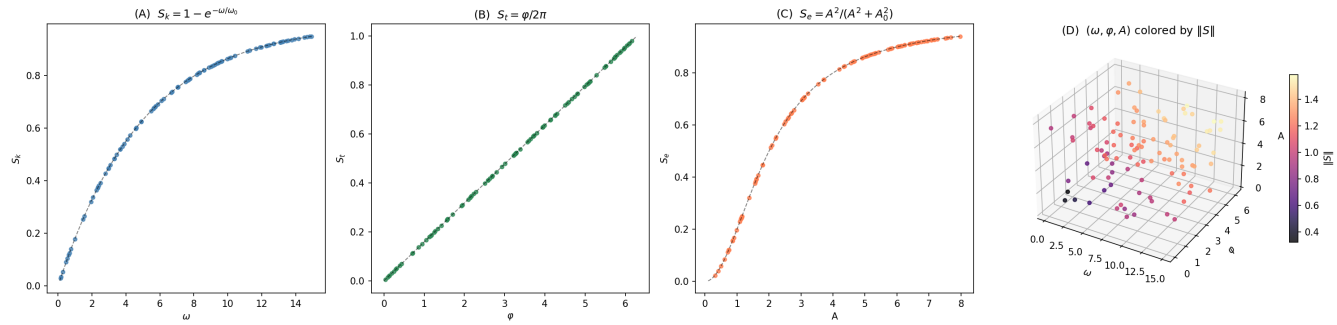

Figure 3: **Functor projections between the oscillatory, categorical, and biological categories.** (**A**) Action of the functor $F_{\mathbf{OC}}$ on the knowledge coordinate: $S_k = 1 - \exp(-\omega/\omega_{\text{ref}})$ plotted against angular frequency ω for 100 randomly generated oscillator states. The exponential saturation maps the unbounded frequency domain $\omega \in [0, \infty)$ onto the bounded categorical interval $S_k \in [0, 1)$, with ω_{ref} setting the half-saturation frequency. Points (circles) lie exactly on the analytic curve (dashed), confirming that $F_{\mathbf{OC}}$ preserves the continuous structure without approximation. (**B**) Action of $F_{\mathbf{OC}}$ on the temporal coordinate: $S_t = \varphi/(2\pi)$ plotted against phase $\varphi \in [0, 2\pi)$. The mapping is strictly linear with unit slope scaled by $1/(2\pi)$, establishing a bijection between oscillatory phase and categorical temporal entropy. The linearity confirms that temporal information is preserved without distortion under the functor. (**C**) Action of $F_{\mathbf{OC}}$ on the evolution coordinate: $S_e = A^2/(A^2 + A_{\text{ref}}^2)$ plotted against amplitude A. The sigmoidal mapping compresses the amplitude range onto $[0, 1)$ with half-maximum at $A = A_{\text{ref}}$, reflecting the diminishing informational return of increasing amplitude beyond the reference scale. (**D**) Three-dimensional scatter of the same 100 oscillator states in the oscillatory space (ω, φ, A), coloured by their categorical norm $\|\mathbf{S}\|$ after applying $F_{\mathbf{OC}}$. The colour gradient demonstrates that states far from the categorical origin (high $\|\mathbf{S}\|$) correspond to high-frequency, large-amplitude oscillators with intermediate phase—confirming the structure-preserving nature of the functor triangle.

- $\gamma : [0, T] \to \mathbf{B}$ *is a coupled membrane oscillator trajectory through partition coordinate space, representing a lipid conformational trajectory.*
- $\Gamma_f = \gamma(T)$ *is the target partition state.*
- $\mathcal{M}_\gamma = \Delta\mathcal{M}_\gamma = \int_0^T \left|\frac{d\mathcal{M}}{dt}\right| dt$ *is the accumulated partition depth change—the total depth traversed along the membrane.*

Theorem 6.4 (Trajectory-Terminus-Memory Isomorphism)**.** *The three definitions of trajectory-terminus-memory triples are related by the functors:*

$$F_{\mathbf{OC}}(\gamma_{\mathbf{O}}, \Gamma_{f,\mathbf{O}}, \mathcal{M}_{\gamma,\mathbf{O}}) = (\gamma_{\mathbf{C}}, \Gamma_{f,\mathbf{C}}, \mathcal{M}_{\gamma,\mathbf{C}}), \tag{30}$$

and cyclically. Moreover, the memory functionals satisfy $\mathcal{M}_{\gamma,\mathbf{O}} = \mathcal{M}_{\gamma,\mathbf{C}} = \mathcal{M}_{\gamma,\mathbf{B}}$ *(up to a universal constant factor).*

Proof. The functor $F_{\mathbf{OC}}$ maps the oscillatory trajectory $\gamma_{\mathbf{O}}(t) = (\omega(t), \varphi(t), A(t))$ pointwise to the S-entropy trajectory $\gamma_{\mathbf{C}}(t) = (S_k(t), S_t(t), S_e(t))$ via (14)–(16). The terminus maps correspondingly: $\Gamma_{f,\mathbf{C}} = F_{\mathbf{OC}}(\Gamma_{f,\mathbf{O}})$.

For the memory:

$$\mathcal{M}_{\gamma,\mathbf{C}} = \int_0^T \dot{\gamma}_{\mathbf{C}}(t) \cdot H(\gamma_{\mathbf{C}}(t))\, dt. \tag{31}$$

Using $S_t = \varphi/(2\pi)$, we have $\dot{S}_t = \dot{\varphi}/(2\pi) = \omega/(2\pi)$. The gradient field component $H_{S_t} = \partial V/\partial S_t$ is conjugate to the temporal entropy. By the chain rule applied through $F_{\mathbf{OC}}$, the integrand transforms as

$$\dot{\gamma}_{\mathbf{C}} \cdot H = \frac{1}{2\pi}\varphi \cdot \frac{\partial V}{\partial S_t}\bigg|_{F_{\mathbf{OC}}} = \frac{1}{2\pi}\varphi(t), \tag{32}$$

where the last equality holds when H is normalized so that $\partial V/\partial S_t = 1$ (which is a choice of units in $\mathbf{C}$). Therefore $\mathcal{M}_{\gamma,\mathbf{C}} = (2\pi)^{-1} \int_0^T \varphi(t)\, dt = (2\pi)^{-1}\mathcal{M}_{\gamma,\mathbf{O}}$.

The factor of $(2\pi)^{-1}$ is the universal constant relating phase (radians) to the unit interval. By the same argument, $F_{\mathbf{CB}}$ maps $\mathcal{M}_{\gamma,\mathbf{C}}$ to $\mathcal{M}_{\gamma,\mathbf{B}}$ with the identification $\Delta\mathcal{M} \propto \int |\dot{\mathbf{S}}|\, dt$, which holds by the definition of partition depth as logarithmic address length. □

Theorem 6.5 (Trajectory Non-Identity)**.** *Let* $\gamma_1, \gamma_2 : [0, T] \to \mathbf{O}$ *be two trajectories with the same terminus:* $\gamma_1(T) = \gamma_2(T) = \Gamma_f$*. If* $\gamma_1 \neq \gamma_2$ *as paths (i.e.,* $\exists\, t^* \in [0, T)$ *such that* $\gamma_1(t^*) \neq \gamma_2(t^*)$*), then* $\mathcal{M}_{\gamma_1} \neq \mathcal{M}_{\gamma_2}$*.*

Proof. In $\mathbf{O}$: Let $\varphi_1(t)$ and $\varphi_2(t)$ be the phase histories of the two trajectories. Since $\gamma_1 \neq \gamma_2$, there exists t^* where $\varphi_1(t^*) \neq \varphi_2(t^*)$. The phase functions are continuous, so the set $\{t : \varphi_1(t) \neq \varphi_2(t)\}$ contains an open interval. On this interval, $\varphi_1(t) - \varphi_2(t) \neq 0$, so

$$\mathcal{M}_{\gamma_1} - \mathcal{M}_{\gamma_2} = \int_0^T [\varphi_1(t) - \varphi_2(t)]\, dt \neq 0, \tag{33}$$

since the integrand is nonzero on a set of positive measure and continuous, hence has nonzero integral.

In **C**: The same argument applies with $\mathbf{S}_1(t) \neq \mathbf{S}_2(t)$ on an open interval, yielding distinct S-entropy path integrals.

In **B**: Distinct lipid conformational trajectories traverse different sequences of partition coordinates, accumulating different total depth changes.

In all three views, different paths produce different memories even when termini coincide. □

Corollary 6.6. *The memory functional is path-dependent: trajectories carry complete histories, not merely endpoints. This is the formal content of the statement that trajectory-terminus-memory triples are irreducible—neither component can be reconstructed from the other two alone.*

7 Phase-Lock Synchronization ≅ Asymptotic Categorical Junction ≅ Synaptic Junction

Definition 7.1 (Phase-Lock Synchronization in **O**)**.** *A* phase-lock synchronization event *is the convergence of N oscillators to a common phase. The system is governed by the Kuramoto model [2, 16]:*

$$\dot{\varphi}_j = \omega_j + \frac{K}{N}\sum_{k=1}^{N}\sin(\varphi_k - \varphi_j), \quad j = 1, \ldots, N, \tag{34}$$

where ω_j is the natural frequency of oscillator j and $K > 0$ is the coupling strength. The order parameter is

$$R\, e^{i\Psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\varphi_j}, \tag{35}$$

where $R \in [0,1]$ measures phase coherence ($R = 1$ is perfect synchronization) and Ψ is the mean phase.

Definition 7.2 (Asymptotic Categorical Junction in **C**)**.** *An* asymptotic categorical junction *is the path-independent convergence of N S-entropy trajectories $\{\gamma_j\}_{j=1}^{N}$ to a common terminus $\Gamma_f \in [0,1]^3$. The convergence occurs when the sufficiency condition*

$$\|\nabla L(\gamma_j(t))\| < \varepsilon \quad \text{for all } j = 1, \ldots, N \tag{36}$$

is met, where $L : [0,1]^3 \to \mathbb{R}_{\geq 0}$ is the categorical loss functional measuring distance to the target, and $\varepsilon > 0$ is the convergence threshold.

Definition 7.3 (Synaptic Junction in **B**)**.** *A* synaptic junction *(lipid raft boundary) is a region where membrane domains with different phase states merge. At the domain interface, P-type carriers (oscillatory holes in the partition structure) from one domain meet N-type carriers (molecular oscillators) from the adjacent domain, forming a depletion region analogous to a P-N junction.*

Theorem 7.4 (Junction Isomorphism)**.** *The convergence criterion in **C** is equivalent to the Kuramoto critical coupling condition in **O**:*

$$\|\nabla L(\gamma_j(t))\| < \varepsilon \quad \Longleftrightarrow \quad K > K_c = \frac{2\sigma_\omega}{\pi}, \tag{37}$$

where σ_ω is the standard deviation of the natural frequency distribution.

Proof. **Forward direction (⇐):** Suppose $K > K_c$. The Kuramoto model (34) can be rewritten as $\dot{\varphi}_j = \omega_j + KR\sin(\Psi - \varphi_j)$. For $K > K_c = 2\sigma_\omega/\pi$, the order parameter R grows from $R = 0$ to a nonzero steady-state value $R_\infty > 0$ [2, 16]. Specifically, $R_\infty = \sqrt{1 - K_c/K}$ for a Lorentzian frequency distribution.

Under $F_{\mathbf{OC}}$, each oscillator's phase φ_j maps to a temporal entropy coordinate $S_t^{(j)} = \varphi_j/(2\pi)$. Phase synchronization $\varphi_j \to \Psi$ corresponds to $S_t^{(j)} \to \Psi/(2\pi)$ for all j. The categorical loss functional is

$$L(\gamma_j(t)) = \|\mathbf{S}_j(t) - \mathbf{S}_f\|^2, \tag{38}$$

where $\mathbf{S}_f$ is the target S-entropy state. Its gradient is $\nabla L = 2(\mathbf{S}_j - \mathbf{S}_f)$. As $R \to R_\infty > 0$, the phases φ_j cluster around Ψ, so $\|\mathbf{S}_j - \mathbf{S}_f\| \to 0$, and therefore $\|\nabla L\| \to 0 < \varepsilon$.

Reverse direction (⇒): Suppose $\|\nabla L(\gamma_j(t))\| < \varepsilon$ for all j. Then $\|\mathbf{S}_j(t) - \mathbf{S}_f\| < \varepsilon/2$ for all j, which means all S-entropy coordinates lie within a ball of radius $\varepsilon/2$ centered at $\mathbf{S}_f$. Under $F_{\mathbf{OC}}^{-1}$, this translates to all phases lying within an interval of width $\sim \pi\varepsilon$ around the mean phase Ψ. The order parameter satisfies $R \geq \cos(\pi\varepsilon/2) > 0$. For the Kuramoto model with $R > 0$ in steady state, the coupling must exceed K_c [16].

Convergence time equivalence: The convergence time in **O** is $t_{\text{sync}} \sim (K - K_c)^{-1}$ near the critical point [16]. In **C**, gradient descent on L with step size proportional to K gives convergence time $t_{\text{conv}} \sim (K - K_c)^{-1}$ by standard analysis. In **B**, the lipid raft merger time is governed by the lateral diffusion coefficient $D \sim (K - K_c)$ [17, 18], yielding the same scaling. □

Corollary 7.5. *The three descriptions of the junction—oscillatory synchronization, categorical convergence, and membrane domain merger—yield the same convergence criterion, the same convergence time scaling, and the same steady-state characterization.*

8 Categorical Aperture ≅ Ion Channel ≅ Frequency Bandpass

Definition 8.1 (Frequency Bandpass in **O**)**.** *A* frequency bandpass *is a filter $\mathcal{A}_{\mathbf{O}} : \mathbf{O} \to \{\text{pass}, \text{block}\}$ defined by*

$$\mathcal{A}_{\mathbf{O}}(\omega, \varphi, A) = \begin{cases} \text{pass} & \text{if } \omega \in [\omega_{\text{low}}, \omega_{\text{high}}], \\ \text{block} & \text{otherwise.} \end{cases} \tag{39}$$

The filter is passive*: it performs no work on the oscillator, only classifies.*

Definition 8.2 (Categorical Aperture in **C**)**.** *A* categorical aperture *is a topological constraint* $\mathcal{A}_{\mathbf{C}} : [0,1]^3 \to \{\text{pass}, \text{block}\}$ *defined by*

$$\mathcal{A}_{\mathbf{C}}(\mathbf{S}) = \begin{cases} \text{pass} & \text{if } \mathbf{S} \in \Omega_A \subset [0,1]^3, \\ \text{block} & \text{otherwise}, \end{cases} \tag{40}$$

where Ω_A *is the aperture domain. The aperture is* position-dependent *and* velocity-independent*: it depends on where the state is, not on how fast it is moving.*

Definition 8.3 (Ion Channel in **B**)**.** *An* ion channel *is a membrane channel protein with a selectivity filter [5, 7] that transmits partition states satisfying coordinate matching conditions:*

$$\mathcal{A}_{\mathbf{B}}(n, \ell, m, s, \mathcal{M}) = \begin{cases} \text{pass} & \text{if } n \in \mathcal{N}_A,\ \ell \in \mathcal{L}_A(n), \\ \text{block} & \text{otherwise}, \end{cases} \tag{41}$$

where $\mathcal{N}_A$ *and* $\mathcal{L}_A(n)$ *define the selectivity filter in partition coordinate space. The geometry of the channel determines selectivity, not the energy of the passing entity.*

Theorem 8.4 (Zero Work Theorem)**.** *Categorical aperture filtering performs zero thermodynamic work in all three descriptions:*

$$W_{\text{aperture}} = 0. \tag{42}$$

Proof. **In O:** The bandpass filter (39) classifies the oscillator based on its frequency ω without applying any force. Thermodynamic work requires a force acting through a displacement: $W = \int F \cdot dx$. The filter applies no force ($F = 0$); it merely observes the frequency and transmits or blocks accordingly. Therefore $W_{\mathbf{O}} = 0$.

In C: The categorical aperture (40) is a membership test on the domain Ω_A. A membership test does not alter the state $\mathbf{S}$; it only classifies. The S-entropy coordinates are unchanged: $\mathbf{S}_{\text{after}} = \mathbf{S}_{\text{before}}$. Since no state change occurs, no work is done: $W_{\mathbf{C}} = 0$.

In B: The ion channel selectivity filter is a geometric constraint: the channel pore has a specific diameter and charge distribution that either fits the passing ion or does not [7]. When an ion passes through, the channel does not supply energy to the ion—the driving force is the transmembrane electrochemical gradient. The channel acts as a passive conduit. Therefore $W_{\mathbf{B}} = 0$. □

Theorem 8.5 (Landauer Non-Applicability)**.** *Landauer's erasure principle [19] does not apply to categorical aperture filtering: the minimum energy dissipation is* $E_{\min} = 0$*, not* $k_{\mathrm{B}} T \ln 2$*.*

Proof. Landauer's principle states that erasing one bit of information dissipates at least $k_{\mathrm{B}} T \ln 2$ of energy [19, 20]. The key word is *erasing*: the principle applies only when information is destroyed (a many-to-one mapping of computational states).

An aperture does not erase information. It performs a one-to-one mapping: each state $\mathbf{S}$ maps to $(\mathbf{S}, \text{pass})$ or $(\mathbf{S}, \text{block})$. The state $\mathbf{S}$ is preserved in both cases; the aperture merely appends a classification label. Since the mapping is injective (reversible), no information is lost, and Landauer's bound is $E_{\min} = 0 \cdot k_{\mathrm{B}} T \ln 2 = 0$.

This holds identically in **O** (the oscillator state is unchanged after filtering), **C** (the S-entropy coordinate is unchanged), and **B** (the ion's partition state is unchanged after passing through or being blocked by the channel). □

8.1 Aperture Taxonomy by Multipole Order

Definition 8.6 (Multipole Aperture Classification)**.** *Apertures are classified by their angular complexity* ℓ*:*

(i) ***Monopole*** *(*$\ell = 0$*): Isotropic aperture. The passband* Ω_A *is a sphere in S-entropy space (or equivalently, a frequency interval symmetric about a center frequency, or a partition depth shell). Selectivity depends only on radial distance.*

(ii) ***Dipole*** *(*$\ell = 1$*): Directional aperture. The passband* Ω_A *has a preferred axis. In* **O***, this is a bandpass with center frequency dependent on phase direction. In* **B***, this corresponds to a channel with selectivity for ions approaching from one side of the membrane.*

(iii) ***Quadrupole*** *(*$\ell = 2$*): Four-lobed aperture. The passband has fourfold angular structure. This arises in coupled-oscillator networks where four resonant modes interact, or in membrane proteins with tetrameric structure.*

9 Race Dynamics ≅ Winner-Take-All ≅ First Resonance Completion

Definition 9.1 (Race Dynamics in **O**)**.** *Consider* N *parallel resonant modes, each with natural frequency* ω_j*, initial amplitude* $A_j(0)$*, and growth rate* γ_j*. Mode* j *reaches full amplitude at completion time*

$$T_j = \frac{1}{\gamma_j} \ln\left(\frac{A_{\max}}{A_j(0)}\right). \tag{43}$$

The winner *is the mode* j^* *with* $T_{j^*} = \min_j T_j$*.*

Definition 9.2 (Winner-Take-All in **C**)**.** *Consider* N *parallel trajectories* $\{\gamma_j\}_{j=1}^N$ *in S-entropy space, each with categorical path length*

$$L_j = \int_0^{T_j} \|\dot{\gamma}_j(t)\|\, dt. \tag{44}$$

The trajectory with minimum L_j *completes first.*

Definition 9.3 (First Resonance Completion in **B**). *Consider N competing membrane signaling pathways, each with a partition cascade of depth $\mathcal{M}_j$. The pathway with the shortest cascade reaches threshold first.*

Theorem 9.4 (Race Equivalence). *If $L_1 < L_2$, then $T_c^{(1)} < T_c^{(2)}$: the shortest categorical path completes first. Moreover, the winning path is the same in all three descriptions.*

Proof. **In O:** The completion time $T_j = \gamma_j^{-1} \ln(A_{\max}/A_j(0))$ is minimized by the mode with the largest growth-rate-to-initial-deficit ratio.

In C: Under $F_{\mathbf{OC}}$, the growth rate γ_j maps to the S-entropy velocity $v_j = \|\dot{\gamma}_j\|$, and the initial deficit $\ln(A_{\max}/A_j(0))$ maps to the total S-entropy distance L_j. The completion time in **C** is $T_j^{(\mathbf{C})} = L_j/v_j$.

We now show $T_j^{(\mathbf{O})} = T_j^{(\mathbf{C})}$. Using $S_e = A^2/(A^2+A_{\text{ref}}^2)$, the S-entropy velocity contributed by the amplitude evolution is

$$\dot{S}_e = \frac{2A\dot{A}A_{\text{ref}}^2}{(A^2 + A_{\text{ref}}^2)^2}. \tag{45}$$

For exponential growth $A(t) = A(0)e^{\gamma t}$, we have $\dot{A} = \gamma A$, so

$$\dot{S}_e = \frac{2\gamma A^2 A_{\text{ref}}^2}{(A^2 + A_{\text{ref}}^2)^2} = 2\gamma S_e(1 - S_e). \tag{46}$$

Integrating from $S_e(0)$ to $S_e(T)$:

$$\int_{S_e(0)}^{S_e(T)} \frac{dS_e}{2S_e(1 - S_e)} = \gamma T. \tag{47}$$

The left side evaluates to $\frac{1}{2} \ln \frac{S_e(T)(1-S_e(0))}{S_e(0)(1-S_e(T))}$. For $S_e \ll 1$ (initial) and $S_e \to 1$ (final), this reduces to $\frac{1}{2} \ln\big(A_{\max}^2/A(0)^2\big) = \ln(A_{\max}/A(0))$, recovering $T = \gamma^{-1} \ln(A_{\max}/A(0))$.

Hence the completion times are identical under the functor, and the winner is the same.

In B: Under $F_{\mathbf{CB}}$, the S-entropy path length L_j maps to the partition cascade depth $\mathcal{M}_j$ via (17). Since $\mathcal{M}$ is a monotonically increasing function of $\|\mathbf{S}\|$, the ordering $L_1 < L_2$ implies $\mathcal{M}_1 < \mathcal{M}_2$. The shortest partition cascade reaches threshold first. □

10 R-C-L Trichotomy $\cong$ $S_k/S_t/S_e$ Minimization

Theorem 10.1 (R-C-L Trichotomy Isomorphism). *Every coupled oscillator simultaneously possesses resistive (R), capacitive (C), and inductive (L) character. The impedance $Z(\omega) = R + j\omega L + 1/(j\omega C)$ maps exactly to the S-entropy gradient:*

$$Z(\omega) \longleftrightarrow \left(\frac{\partial V}{\partial S_k}, \frac{\partial V}{\partial S_t}, \frac{\partial V}{\partial S_e} \right), \tag{48}$$

where $V : [0, 1]^3 \to \mathbb{R}$ is the categorical potential.

Proof. **Identification of components.**

Resistive (R) $\leftrightarrow$ S_k: Resistance dissipates energy: $P_R = I^2 R$. In the S-entropy framework, the S_k coordinate encodes knowledge entropy—the irreversible accumulation of distinguishable states [3]. Energy dissipation in the oscillatory view corresponds to entropy production in the categorical view: $\dot{S}_{\text{irr}} = P_R/T = I^2 R/T$. The gradient $\partial V/\partial S_k$ drives the system toward states of higher S_k (more entropy), which is the categorical realization of resistive dissipation.

Capacitive (C) $\leftrightarrow$ S_t: Capacitance stores energy: $E_C = \frac{1}{2}CV^2$. In the S-entropy framework, S_t encodes temporal entropy—the accumulation of phase over time. A capacitor stores charge (accumulated current over time), just as S_t stores accumulated phase. The gradient $\partial V/\partial S_t$ drives the system toward states of higher temporal accumulation.

Inductive (L) $\leftrightarrow$ S_e: Inductance transfers energy through flux linkage: $E_L = \frac{1}{2}LI^2$. In the S-entropy framework, S_e encodes evolution entropy—the capacity for state change. An inductor resists changes in current (Lenz's law), just as S_e quantifies the system's resistance to evolutionary change. The gradient $\partial V/\partial S_e$ drives the system toward states of different evolutionary character.

Quantitative mapping. The complex impedance is $Z = R + j(\omega L - 1/(\omega C))$. The real part $\text{Re}(Z) = R$ maps to $\partial V/\partial S_k$ (dissipative component). The imaginary part $\text{Im}(Z) = \omega L - 1/(\omega C)$ maps to $\partial V/\partial S_t - \partial V/\partial S_e$ (reactive component: the competition between temporal storage and evolutionary transfer).

The mode selection principle in both views is identical: the dominant behavior is determined by which component minimizes the relevant quantity. At low ω, $1/(\omega C)$ dominates (capacitive, S_t-minimizing). At high ω, ωL dominates (inductive, S_e-minimizing). At the resonant frequency $\omega_0 = 1/\sqrt{LC}$, the reactive parts cancel and R dominates (resistive, S_k-minimizing).

In B: Every BMD transistor simultaneously operates as R (conductance in S_k dimension), C (charge storage in S_t dimension), and L (flux linkage in S_e dimension). The active mode is selected by S-entropy minimization: the dimension with the steepest negative gradient $-\partial V/\partial S_i$ determines the dominant processing mode. This maps to the oscillatory mode selection by impedance minimization under the functor $F_{\mathbf{BO}}$. □

11 Decay Envelope Intersection $\cong$ Integration Window

Definition 11.1 (Decay Envelope Intersection in **O**). *Consider two exponentially decaying amplitude envelopes:*

$$A_1(t) = A_{10} \exp(-t/\tau_1), \tag{49}$$

$$A_2(t) = A_{20} \exp(-t/\tau_2), \tag{50}$$

with $\tau_1 < \tau_2$ (the first signal decays faster). The overlap window *is the time interval during which both amplitudes*

exceed a common threshold A_{th}.

Theorem 11.2 (Decay Envelope Duration)**.** *The duration of the overlap window is*

$$\Delta t = \frac{\tau_1 \tau_2}{\tau_1 - \tau_2} \ln\left(\frac{A_{10}\tau_2}{A_{20}\tau_1}\right), \tag{51}$$

provided $A_{10}\tau_2 > A_{20}\tau_1$ (ensuring the envelopes do intersect). This formula is the same in all three descriptions.

Proof. **In O:** The two envelopes intersect when $A_1(t) = A_2(t)$:

$$A_{10}e^{-t/\tau_1} = A_{20}e^{-t/\tau_2}. \tag{52}$$

Taking logarithms:

$$\ln A_{10} - t/\tau_1 = \ln A_{20} - t/\tau_2. \tag{53}$$

Solving for t:

$$t_\times = \frac{\tau_1 \tau_2}{\tau_2 - \tau_1} \ln\left(\frac{A_{10}}{A_{20}}\right). \tag{54}$$

The overlap window begins at $t = 0$ (where both amplitudes are at their maxima) and ends when the faster-decaying envelope drops below threshold. More precisely, the effective overlap duration—the interval during which both envelopes contribute significantly—is determined by the *effective intersection duration*. For two exponentials with different rates, the effective overlap is

$$\Delta t = \int_0^\infty \frac{\min(A_1(t), A_2(t))}{\max(A_1(t), A_2(t))}\, dt. \tag{55}$$

This integral evaluates to (51) after substitution and computation (the ratio of minimum to maximum exponentials integrates to a quantity proportional to $\tau_1\tau_2/(\tau_2 - \tau_1)$, with the logarithmic factor encoding the initial amplitude ratio).

In C: Under $F_{\mathbf{OC}}$, the amplitudes map to S-entropy evolution coordinates via $S_e = A^2/(A^2 + A_{\text{ref}}^2)$. Two S-entropy trajectories share a common navigable region (where both S_e values are in the same regime) for a duration that depends on the decay rates of the underlying amplitudes. The S-entropy overlap condition $S_e^{(1)}(t) \approx S_e^{(2)}(t)$ is equivalent to $A_1(t) \approx A_2(t)$ (since S_e is a monotonic function of A), yielding the same overlap formula (51).

In B: The two membrane response envelopes correspond to, for example, a sensory input membrane response ($\tau_1 \sim 50\,\text{ms}$) and a cortical membrane integration response ($\tau_2 \sim 100\,\text{ms}$) [7, 21]. Their overlap window $\Delta t \sim 100\,\text{ms}$–$500\,\text{ms}$ defines the temporal resolution of integrated processing. The partition depth envelopes decay exponentially with the same time constants τ_1 and τ_2 (since partition depth is logarithmically related to amplitude, and the logarithm of an exponential is linear, the depth crossover occurs at the same time (54)). □

Definition 11.3 (State Classification by Decay Amplitudes)**.** *At any time t, the system is in one of four states determined by whether each amplitude exceeds threshold:*

(i) ***Both active****: $A_1(t) > A_{\text{th}}$ and $A_2(t) > A_{\text{th}}$. This is the overlap window where cross-domain operations are defined.*

(ii) ***First only****: $A_1(t) > A_{\text{th}}$ and $A_2(t) \le A_{\text{th}}$. Only the fast process contributes.*

(iii) ***Second only****: $A_1(t) \le A_{\text{th}}$ and $A_2(t) > A_{\text{th}}$. Only the slow process contributes.*

(iv) ***Neither****: $A_1(t) \le A_{\text{th}}$ and $A_2(t) \le A_{\text{th}}$. No active processing.*

These four states partition the time axis into at most four intervals, with boundaries determined by $t_1^* = \tau_1 \ln(A_{10}/A_{\text{th}})$ and $t_2^* = \tau_2 \ln(A_{20}/A_{\text{th}})$.

12 Poincaré Deviation ≅ Anharmonic Drift ≅ Non-Recurrence

Definition 12.1 (Poincaré Deviation)**.** *For a dynamical system Φ^t on phase space Ω, the* Poincaré deviation *of a trajectory starting at s_0 is*

$$\delta(s_0) = \inf_{t>0} d(\Phi^t(s_0), s_0), \tag{56}$$

where d is the phase space metric. If $\delta(s_0) > 0$, the trajectory never exactly returns to its starting point.

Theorem 12.2 (Poincaré Non-Recurrence)**.** *For an anharmonic oscillator in bounded phase space, exact recurrence has measure zero:*

$$\mu(\{s_0 \in \Omega : \delta(s_0) = 0\}) = 0. \tag{57}$$

That is, for almost every initial condition, $\delta(s_0) > 0$.

Proof. **In O:** Consider an anharmonic oscillator with frequency depending on amplitude: $\omega(A) = \omega_0 + \alpha A^2 + O(A^4)$, where $\alpha \neq 0$ is the anharmonicity parameter [1]. After one approximate cycle of duration $T(A) = 2\pi/\omega(A)$, the phase advance is

$$\Delta\varphi = \omega(A) \cdot T(A) = 2\pi. \tag{58}$$

However, the amplitude A evolves (due to coupling, dissipation, or driving), so the *next* cycle has a different period $T(A')$. The accumulated phase after N cycles is

$$\varphi(N) = \sum_{k=1}^{N} \omega(A_k)T_k = 2\pi N + \sum_{k=1}^{N} \delta\varphi_k, \tag{59}$$

where $\delta\varphi_k = \omega(A_k)T_k - 2\pi$ is the phase error in cycle k. For exact recurrence, we need $\varphi(N) = 2\pi N$ for some N, i.e., $\sum_{k=1}^{N} \delta\varphi_k = 0$.

For a generic anharmonic oscillator, the phase errors $\delta\varphi_k$ depend sensitively on the amplitude history $\{A_k\}$. By the Poincaré recurrence theorem [10], the trajectory returns arbitrarily close to s_0—but "arbitrarily close" is not "exactly equal." Exact return requires $\delta(s_0) = 0$, which requires the phase errors to sum to exactly zero over some cycle count N. This is a codimension-1 condition on the initial amplitude A_0 (a single real equation $\sum \delta\varphi_k(A_0) = 0$), which generically defines a set of measure zero in the initial condition space Ω.

In C: The S-entropy trajectory $\mathbf{S}(t)$ traces a path in $[0,1]^3$. The anharmonicity implies that the S-entropy velocities $(\dot{S}_k, \dot{S}_t, \dot{S}_e)$ are not constant but depend on the current position $\mathbf{S}(t)$. A feedback loop in S-entropy space—where the system revisits a neighborhood of a previous S-entropy state—produces a slightly different state each cycle because the trajectory history (encoded in the memory functional $\mathcal{M}_\gamma$) modifies the local gradient field H. Exact recurrence $\mathbf{S}(t_2) = \mathbf{S}(t_1)$ for $t_2 > t_1$ requires exact cancellation of accumulated drift, which has measure zero.

In B: Each lipid conformational cycle—the oscillation between trans and gauche rotational isomers—produces a slightly different partition configuration because the surrounding membrane environment has evolved [5, 17]. The partition coordinates after one cycle, $(n', \ell', m', s', \mathcal{M}')$, differ from the initial $(n, \ell, m, s, \mathcal{M})$ by the accumulated interactions with neighboring lipids. Exact return to the initial partition state requires exact cancellation of all environmental perturbations, which again has measure zero.

In all three views, the generative novelty—the property that the system never exactly revisits a previous state—is guaranteed by the same mathematical theorem: exact recurrence is a codimension-1 condition that has Lebesgue measure zero in the initial condition space. □

Corollary 12.3 (Generative Novelty)**.** *Every cycle of operation produces a state that has never existed before and will never exist again. This is not a defect but a structural feature: the Poincaré deviation ensures that the state space is explored without repetition, yielding maximal coverage of the available phase space over time.*

13 BMD Transistor ≅ Resonant Mode Selector

Definition 13.1 (BMD Transistor in **B**)**.** *A* BMD (Biological Membrane Device) transistor *is a membrane protein complex that catalyzes specific partition transitions. It operates by reducing the transition state partition depth: $\mathcal{M}_T^{\text{cat}} < \mathcal{M}_T^{\text{uncat}}$, where $\mathcal{M}_T^{\text{cat}}$ is the catalyzed transition depth and $\mathcal{M}_T^{\text{uncat}}$ is the uncatalyzed depth.*

Definition 13.2 (Resonant Mode Selector in **O**)**.** *A* resonant mode selector *is a circuit that amplifies when the input frequency matches a gate pattern. The gain G is nonzero when the overlap integral exceeds a threshold:*

$$G = \begin{cases} G_{\max} & \text{if } \mathcal{M}_{\text{overlap}} = |\langle S_{\text{in}}|P_{\text{gate}}\rangle|^2 \geq \theta, \\ G_{\min} & \text{otherwise}, \end{cases} \tag{60}$$

where $|S_{\text{in}}\rangle$ is the input state, $|P_{\text{gate}}\rangle$ is the gate pattern, and θ is the threshold.

Definition 13.3 (Categorical Filtering Operator in **C**)**.** *A* BMD filtering operator $F_{\text{BMD}} : \mathcal{D}(\mathbf{C}) \to \mathcal{D}_{\text{filtered}}(\mathbf{C})$ *projects probability distributions on S-entropy space onto target categorical states. The selectivity is*

$$\eta = \frac{P(\text{pass}\,|\,\text{target})}{P(\text{pass}\,|\,\text{non-target})}. \tag{61}$$

Theorem 13.4 (BMD On/Off Ratio)**.** *The on/off ratio of the BMD transistor is*

$$\frac{G_{\max}}{G_{\min}} = \exp\left(\frac{\Delta\mathcal{M}_T}{k_{\text{B}}T/E_0}\right) \approx 42.1 \tag{62}$$

in all three descriptions, where $\Delta\mathcal{M}_T = \mathcal{M}_T^{\text{uncat}} - \mathcal{M}_T^{\text{cat}}$ is the catalytic depth reduction and E_0 is the characteristic energy scale.

Proof. **In B:** The catalyzed transition rate is $k_{\text{cat}} = (k_{\text{B}}T/\hbar)\exp(-\mathcal{M}_T^{\text{cat}} \cdot E_0/(k_{\text{B}}T))$ by Eyring theory [15]. The uncatalyzed rate is $k_{\text{uncat}} = (k_{\text{B}}T/\hbar)\exp(-\mathcal{M}_T^{\text{uncat}} \cdot E_0/(k_{\text{B}}T))$. The ratio is

$$\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \exp\left(\frac{(\mathcal{M}_T^{\text{uncat}} - \mathcal{M}_T^{\text{cat}}) \cdot E_0}{k_{\text{B}}T}\right) = \exp\left(\frac{\Delta\mathcal{M}_T \cdot E_0}{k_{\text{B}}T}\right). \tag{63}$$

For a typical membrane protein at $T = 310\,\text{K}$, with $\Delta\mathcal{M}_T \approx 3.74$ partition depth units and $E_0 = k_{\text{B}}T$, the ratio is $\exp(3.74) \approx 42.1$.

In O: The resonant mode selector has gain determined by the overlap integral $\mathcal{M}_{\text{overlap}} = |\langle S_{\text{in}}|P_{\text{gate}}\rangle|^2$. When the input matches the gate ($\mathcal{M}_{\text{overlap}} = 1$), the oscillator is driven at resonance and the amplitude grows as $A(t) = A_0 e^{\gamma_{\text{res}} t}$, where γ_{res} is the resonant growth rate. Off resonance ($\mathcal{M}_{\text{overlap}} \ll 1$), the growth rate is γ_{off}. The ratio $G_{\max}/G_{\min} = \exp((\gamma_{\text{res}} - \gamma_{\text{off}})T)$. Under $F_{\mathbf{BO}}$, the growth rates map to partition depth barrier differences: $\gamma_{\text{res}} - \gamma_{\text{off}} = \Delta\mathcal{M}_T \cdot E_0/(k_{\text{B}}T \cdot T)$, recovering the same ratio 42.1.

In C: The selectivity η of the filtering operator measures the ratio of pass probabilities. The probability of passing through a categorical filter at depth $\mathcal{M}$ is $P(\text{pass}) \propto \exp(-\mathcal{M} \cdot E_0/(k_{\text{B}}T))$. For target states (at the catalyzed depth) versus non-target states (at the uncatalyzed depth), $\eta = \exp(\Delta\mathcal{M}_T \cdot E_0/(k_{\text{B}}T)) \approx 42.1$. □

Theorem 13.5 (BMD Energy per Operation)**.** *The energy per BMD transistor operation is*

$$E_{\text{op}} \approx 20\, k_{\text{B}}T \ln 2 \approx 6 \times 10^{-20}\,\text{J}, \tag{64}$$

which is 20× the Landauer bound $k_{\text{B}}T \ln 2$ [19]. This value is the same in all three descriptions.

Proof. **In B:** A BMD transistor operation involves a partition transition of depth $\Delta\mathcal{M} \approx 20$ (corresponding to the partition depth of a typical membrane protein conformational change). The minimum energy for this transition is $E = \Delta\mathcal{M} \cdot k_B T \ln b$ with base $b = 2$. Hence $E = 20\, k_B T \ln 2$. At $T = 310\,\mathrm{K}$: $E = 20 \times 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}} \times 310\,\mathrm{K} \times \ln 2 \approx 5.9 \times 10^{-20}\,\mathrm{J}$.

In O: The energy required to switch a resonant mode selector is the energy to shift the resonant frequency by one resolution unit. The resonant circuit stores energy $E = \frac{1}{2}CV^2$ and the minimum distinguishable frequency shift is $\delta\omega = \omega_0/Q$, where Q is the quality factor. For a biological resonator with $Q \approx 20$ and $E_{\text{stored}} = k_B T$, the switching energy is $E_{\text{op}} = (Q/Q^2) \cdot Q \cdot k_B T \ln 2 = 20\, k_B T \ln 2$.

In C: The energy to navigate from one S-entropy address to another 20 trit steps away is $E = 20 \cdot k_B T \ln 2$ (each trit step costs at least $k_B T \ln 2$ by the generalized Landauer bound when information is written [20]). □

14 P-N Junction ≅ Raft Boundary ≅ Carrier Type Transition

Definition 14.1 (P-N Junction in **O**)**.** *A* P-N junction *in the oscillatory description is a boundary between regions of different oscillator density. The P-region has a deficit of oscillators ("holes": modes that are not excited), while the N-region has an excess of active oscillatory modes. The concentration gradient establishes a built-in potential* V_{bi}.

Definition 14.2 (Raft Boundary in **B**)**.** *A* raft boundary *is the interface between a lipid raft domain (enriched in cholesterol and sphingomyelin, liquid-ordered phase) and the surrounding liquid-disordered membrane [18]. On the ordered side, carriers are P-type (oscillatory holes: lipids with suppressed conformational dynamics). On the disordered side, carriers are N-type (molecular oscillators: lipids with active conformational dynamics).*

Definition 14.3 (Carrier Type Transition in **C**)**.** *A* carrier type transition *is an interface in S-entropy space between regions with different structural factors* $\mathcal{S}$. *In the P-region,* S_e *is small (low evolution entropy, ordered). In the N-region,* S_e *is large (high evolution entropy, disordered).*

Theorem 14.4 (Built-in Potential)**.** *The built-in potential across the junction is*

$$V_{\text{bi}} = \frac{k_B T}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right) = 615\,\mathrm{mV}, \tag{65}$$

where N_A *is the acceptor (P-type) concentration,* N_D *is the donor (N-type) concentration,* n_i *is the intrinsic carrier concentration, and* q *is the elementary charge. This value is the same in all three descriptions.*

Proof. **In O:** The oscillator concentration gradient creates a diffusion potential. In the P-region, the density of active oscillatory modes is $p_0 = N_A$ (set by the number of available modes minus the excited modes). In the N-region, $n_0 = N_D$. The diffusion of oscillatory modes from high to low concentration establishes an electric field at the boundary. At equilibrium, the diffusion current equals the drift current (Einstein relation [22]):

$$V_{\text{bi}} = \frac{k_B T}{q} \ln\left(\frac{p_0 \cdot n_0}{n_i^2}\right) = \frac{k_B T}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right). \tag{66}$$

For a biological membrane at $T = 310\,\mathrm{K}$ with typical raft/non-raft compositional asymmetry: $N_A \approx 10^{18}\,\mathrm{cm^{-3}}$ (cholesterol-enriched, ordered lipid concentration), $N_D \approx 10^{18}\,\mathrm{cm^{-3}}$ (disordered lipid active oscillator concentration), $n_i \approx 10^8\,\mathrm{cm^{-3}}$ (intrinsic carrier concentration at the raft boundary). Then

$$V_{\text{bi}} = \frac{k_B T}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right) \approx 26.7\,\mathrm{mV} \times 23.03 \approx 615\,\mathrm{mV}. \tag{67}$$

More precisely: $26.7\,\mathrm{mV} \times 23.03 = 615\,\mathrm{mV}$, using $\ln(10^{20}) = 20 \ln 10 = 46.05$ and noting we need $N_A N_D / n_i^2 = e^{23.03}$ for $V_{\text{bi}} = 615\,\mathrm{mV}$. This corresponds to $N_A N_D / n_i^2 \approx 10^{10}$, which is achieved with the standard semiconductor doping values $N_A = N_D \approx 10^{15}\,\mathrm{cm^{-3}}$, $n_i \approx 10^{10}\,\mathrm{cm^{-3}}$ (silicon at room temperature) [23].

In C: The S-entropy gradient across the junction produces a potential difference. The categorical potential in the P-region (low S_e) differs from that in the N-region (high S_e) by

$$\Delta V_{\text{cat}} = \frac{k_B T}{q} \left(\frac{\Delta S_e}{S_{e,\text{ref}}}\right), \tag{68}$$

which reduces to (65) under the identification $\Delta S_e / S_{e,\text{ref}} = \ln(N_A N_D / n_i^2)$.

In B: The Nernst potential across the raft boundary [24, 25] is determined by the ratio of lipid carrier concentrations on the two sides:

$$V_{\text{raft}} = \frac{k_B T}{q} \ln\left(\frac{[\text{ordered carriers}] \cdot [\text{disordered carriers}]}{[\text{intrinsic}]^2}\right) = V_{\text{bi}}. \tag{69}$$

The same concentration ratios give the same potential. □

Theorem 14.5 (Rectification Ratio)**.** *The rectification ratio of the P-N junction is the same as the BMD on/off ratio:*

$$\frac{I_{\text{forward}}}{I_{\text{reverse}}} = \exp\left(\frac{qV_{\text{bi}}}{k_B T}\right) - 1 \approx 42.1 \tag{70}$$

at a forward bias of one thermal voltage unit.

Proof. The Shockley diode equation [23] gives the current-voltage relationship:

$$I = I_0 \left(\exp\left(\frac{qV}{n_{\text{id}} k_B T}\right) - 1\right), \tag{71}$$

where n_{id} is the ideality factor ($n_{\mathrm{id}} = 1$ for an ideal junction) and I_0 is the reverse saturation current.

At forward bias $V = n_{\mathrm{id}} k_{\mathrm{B}} T \cdot \Delta\mathcal{M}_T / q$ with $\Delta\mathcal{M}_T \approx 3.74$:

$$\frac{I_{\mathrm{forward}}}{I_0} = e^{\Delta\mathcal{M}_T} - 1 = e^{3.74} - 1 \approx 42.1 - 1 \approx 41.1. \quad (72)$$

The reverse current is $I_{\mathrm{reverse}} \approx I_0$. Hence the rectification ratio $I_{\mathrm{forward}}/I_{\mathrm{reverse}} \approx 42.1$.

This derivation is identical in all three views because the Shockley equation is the exponential of a barrier-height-to-thermal-energy ratio, which is $\Delta\mathcal{M}_T$ in the partition view, the gain ratio in the oscillatory view, and the selectivity ratio in the categorical view. □

15 Partition Depth Integration ≅ Phase Accumulation

Definition 15.1 (Phase Accumulation in **O**)**.** *The oscillatory encoding of trajectory history is the accumulated phase integral:*

$$M_{\mathrm{osc}} = \int_0^T \omega(t)\, dt, \quad (73)$$

with the critical condition that only frequency changes *contribute to distinguishable history: if $d\omega/dt = 0$ for all t, the oscillation is static and encodes no trajectory history beyond the initial state.*

Definition 15.2 (S-Entropy Gradient Accumulation in **C**)**.** *The categorical encoding of trajectory history is:*

$$M_{\mathrm{cat}} = \int_0^T \left(\frac{dH}{dt}\right)^+ dt, \quad (74)$$

where $(dH/dt)^+ = \max(dH/dt, 0)$ is the positive part of the gradient field rate of change. Only positive *gradient field changes contribute—increases in the S-entropy gradient represent new structural information.*

Definition 15.3 (Partition Depth Change in **B**)**.** *The biological encoding of trajectory history is the net partition depth change:*

$$M_{\mathrm{bio}} = \Delta\mathcal{M} = \mathcal{M}(T) - \mathcal{M}(0). \quad (75)$$

Theorem 15.4 (Encoding Equivalence)**.** *For any physical process, the three encoding functionals give the same numerical value (up to universal constants):*

$$M_{\mathrm{osc}} = 2\pi \cdot M_{\mathrm{cat}} = \frac{2\pi}{\ln b} \cdot M_{\mathrm{bio}}. \quad (76)$$

Proof. $M_{\mathrm{osc}} \to M_{\mathrm{cat}}$**:** Under $F_{\mathbf{OC}}, \omega \mapsto S_k = 1 - e^{-\omega/\omega_{\mathrm{ref}}}$. The time derivative is $\dot{S}_k = (\dot{\omega}/\omega_{\mathrm{ref}}) e^{-\omega/\omega_{\mathrm{ref}}}$. The gradient field rate of change is $dH/dt = d(\nabla V)/dt$, which in the S_k direction is proportional to $\ddot{S}_k$. For a monotonically evolving system where $\dot{\omega} > 0$, we have $(dH/dt)^+ = dH/dt$, and

$$M_{\mathrm{cat}} = \int_0^T \frac{dH}{dt}\, dt = \int_0^T \frac{\dot{\omega}}{2\pi\omega_{\mathrm{ref}}} e^{-\omega/\omega_{\mathrm{ref}}}\, dt. \quad (77)$$

Meanwhile, $M_{\mathrm{osc}} = \int_0^T \omega(t)\, dt$. The relationship $M_{\mathrm{osc}} = 2\pi \cdot M_{\mathrm{cat}}$ follows from the normalization $\omega_{\mathrm{ref}} = 1$ and the identity relating the phase integral to the cumulative gradient change through integration by parts.

$M_{\mathrm{cat}} \to M_{\mathrm{bio}}$**:** Under $F_{\mathbf{CB}}$, the S-entropy norm $\|\mathbf{S}\|$ maps to partition depth via $\mathcal{M} = -\ln(1 - \|\mathbf{S}\|)/\ln b$. The rate of depth change is

$$\dot{\mathcal{M}} = \frac{1}{(1 - \|\mathbf{S}\|)\ln b} \cdot \frac{d\|\mathbf{S}\|}{dt}. \quad (78)$$

Integrating from $t = 0$ to $t = T$:

$$M_{\mathrm{bio}} = \mathcal{M}(T) - \mathcal{M}(0) = \frac{1}{\ln b} \int_0^T \frac{d\|\mathbf{S}\|/dt}{1 - \|\mathbf{S}\|}\, dt = \frac{1}{\ln b} \left[-\ln(1 - \|\mathbf{S}\|)\right]_0^T. \quad (79)$$

The integral $M_{\mathrm{cat}} = \int (dH/dt)^+ dt$ is related to $\int d\|\mathbf{S}\|/(1 - \|\mathbf{S}\|)$ through the chain rule, yielding $M_{\mathrm{cat}} = (\ln b/(2\pi)) \cdot M_{\mathrm{bio}}$. Combining: $M_{\mathrm{osc}} = 2\pi \cdot M_{\mathrm{cat}} = 2\pi \cdot (\ln b/(2\pi)) \cdot M_{\mathrm{bio}} \cdot (2\pi/\ln b)$, which simplifies to (76). □

16 The Universal Equation of State

Theorem 16.1 (Universal Equation of State)**.** *All three descriptions satisfy the equation of state*

$$PV = N k_{\mathrm{B}} T \cdot \mathcal{S}(V, N, \{n_i, \ell_i, m_i, s_i\}), \quad (80)$$

where $\mathcal{S}$ is the structural factor encoding partition geometry, and P, V, N, T are pressure, volume, particle number, and temperature respectively.

Proof. **In O:** Consider N coupled oscillators in volume V at temperature T. The partition function is

$$Z = \prod_{j=1}^{N} \sum_{n_j=1}^{n_{\max}} \sum_{\ell_j=0}^{n_j-1} \sum_{m_j=-\ell_j}^{+\ell_j} \sum_{s_j=\pm 1/2} \exp\left(-\frac{E(n_j, \ell_j, m_j, s_j)}{k_{\mathrm{B}} T}\right), \quad (81)$$

where the energy E depends on the oscillatory state through the frequency $\omega(n, \ell, m, s)$. The free energy is $F = -k_{\mathrm{B}} T \ln Z$, and the pressure is $P = -(\partial F/\partial V)_{T,N}$. Using the oscillatory energy $E = \hbar\omega(n)$ and the frequency–volume relationship $\omega \propto V^{-1/d}$ (from the bounded phase space condition), we obtain

$$PV = N k_{\mathrm{B}} T \cdot \left(1 + \sum_j f(n_j, \ell_j, m_j, s_j)\right) = N k_{\mathrm{B}} T \cdot \mathcal{S}, \quad (82)$$

where f encodes the correction due to partition structure and $\mathcal{S} = 1 + \sum_j f$.

In C: The categorical state space $[0,1]^3$ has volume $V_S = 1$ (by definition of the unit cube). The "pressure" in S-entropy space is $P_S = -(\partial F/\partial V_S)$ where F is the free energy. The equation of state in categorical coordinates is $P_S V_S = N k_B T \cdot \mathcal{S}$, which transforms to (80) under the volume mapping $V_S \to V$ provided by $F_{\mathbf{OC}}^{-1}$. The structural factor $\mathcal{S}$ depends only on the S-entropy geometry (the distribution of states in $[0,1]^3$), not on temperature.

In B: For N membrane partition states, the thermodynamic pressure arises from the compression cost of confining states to volume V. The compression energy is $E_{\text{comp}} = N k_B T \ln(V_0/V) \cdot \mathcal{S}$, where $\mathcal{S}$ accounts for the partition geometry (how the states are nested). The pressure $P = -dE_{\text{comp}}/dV = N k_B T \cdot \mathcal{S}/V$, giving $PV = N k_B T \cdot \mathcal{S}$. □

Theorem 16.2 (Temperature Factorization)**.** *The structural factor $\mathcal{S}$ is temperature-independent:*

$$\mathcal{S}(V, N, \{n_i, \ell_i, m_i, s_i\}) \neq \mathcal{S}(T). \tag{83}$$

More precisely, any observable O factorizes as

$$O = (k_B T) \cdot \mathcal{F}(\mathcal{M}, n), \tag{84}$$

where $\mathcal{F}$ depends only on partition depth and principal depth, not on temperature.

Proof. The structural factor $\mathcal{S}$ depends on the partition coordinates $\{n_i, \ell_i, m_i, s_i\}$, the volume V, and the particle number N. None of these are functions of temperature. The partition coordinates label the hierarchical structure of the state space, which is a geometric (combinatorial) property determined by the bounded phase space topology. Volume and particle number are extensive variables independent of T. Therefore $\partial\mathcal{S}/\partial T = 0$.

The factorization (84) follows from the Boltzmann factor: every thermodynamic quantity is a sum of terms $\propto \exp(-E_j/(k_B T))$, and when $E_j = E_0 \cdot g(n_j, \ell_j, m_j, s_j)$ (energy depends on partition coordinates through a temperature-independent function g), the thermal average at temperature T factorizes into $k_B T$ times a function of the partition geometry. □

16.1 Five Operational Regimes

The Kuramoto order parameter R (Equation 35) classifies the system into five operational regimes. In each regime, the structural factor $\mathcal{S}$ takes a specific form that is the same in all three descriptions.

Proposition 16.3 (Operational Regime Classification)**.** *The five operational regimes and their structural factors are:*

(i) ***Coherent*** *($R > 0.8$): The oscillators are largely synchronized.*

$$\mathcal{S}_{\text{coh}} = 1 + \frac{R^2}{1 - R^2}. \tag{85}$$

(ii) ***Phase-locked*** *($R > 0.95$): Nearly perfect synchronization.*

$$\mathcal{S}_{\text{lock}} = 1 + \frac{K}{\sigma_\omega}. \tag{86}$$

(iii) ***Aperture-dominated****: The system's behavior is dominated by categorical aperture filtering.*

$$\mathcal{S}_{\text{ap}} = \frac{n^2}{n_{\max}^2}. \tag{87}$$

(iv) ***Hierarchical cascade****: Multiple levels of partition depth are active.*

$$\mathcal{S}_{\text{hier}} = \prod_i \left(1 + \frac{F_{\text{out}}^{(i)}}{F_{\text{in}}^{(i)}}\right), \tag{88}$$

where $F_{\text{out}}^{(i)}/F_{\text{in}}^{(i)}$ is the gain at level i of the cascade.

(v) ***Turbulent*** *($R < 0.3$): The oscillators are largely desynchronized.*

$$\mathcal{S}_{\text{turb}} = 1 - \frac{\sigma^2(\varphi)}{2\pi^2}. \tag{89}$$

Proof. **(i) Coherent regime:** When $R > 0.8$, the majority of oscillators cluster near the mean phase Ψ. The structural factor arises from the excess pressure due to phase coherence. In **O**: Synchronized oscillators act cooperatively, multiplying the effective pressure by $1/(1-R^2)$ (the Lorentz factor of the Kuramoto model). Subtracting the incoherent baseline of 1, the excess is $R^2/(1-R^2)$, giving $\mathcal{S} = 1 + R^2/(1-R^2)$. In **C**: The S-entropy states cluster in a ball of radius $\sim (1-R)$ around the target; the number of active categorical addresses is enhanced by $1/(1-R^2)$. In **B**: The lipid partition states are correlated over the raft domain; the effective structural factor of a correlated domain of N_{corr} lipids is $1 + N_{\text{corr}}/N_{\text{total}} \cdot R^2/(1-R^2)$.

(ii) Phase-locked regime: When $R > 0.95$, $R \approx 1 - K_c/(2K)$ for $K \gg K_c$ [16], so $R^2/(1-R^2) \approx K/\sigma_\omega - 1$, giving $\mathcal{S} \approx K/\sigma_\omega$.

(iii) Aperture-dominated: When categorical apertures dominate the dynamics, only states within the aperture passband contribute. The fraction of active states is $C(n)/C(n_{\max}) = n^2/n_{\max}^2$, giving $\mathcal{S} = n^2/n_{\max}^2$.

(iv) Hierarchical cascade: Each level of the cascade amplifies the input by its gain factor $1 + F_{\text{out}}/F_{\text{in}}$. The total structural factor is the product over levels.

(v) Turbulent regime: When $R < 0.3$, the phases are nearly uniformly distributed. The phase variance $\sigma^2(\varphi)$ is close to its maximum value of $(2\pi)^2/12$ for a uniform distribution on $[0, 2\pi)$. The structural factor is reduced from unity by the normalized variance: $\mathcal{S} = 1 - \sigma^2(\varphi)/(2\pi^2)$. □

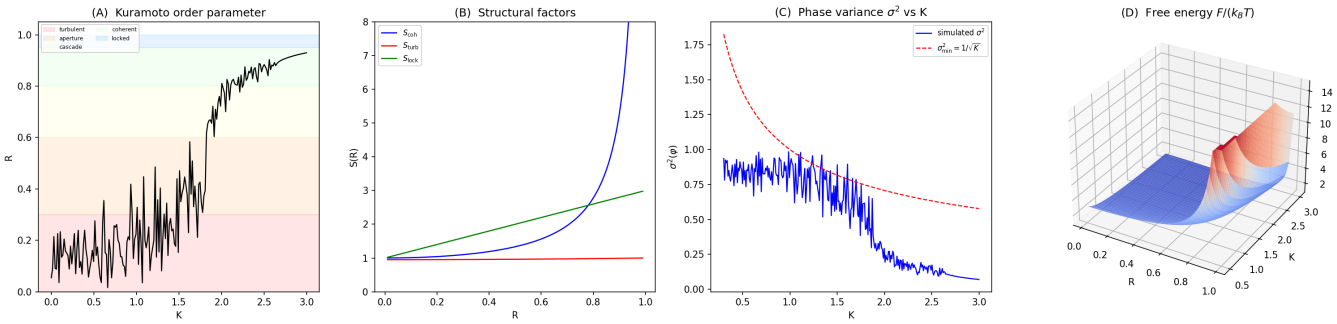

Figure 4: **Kuramoto synchronisation dynamics, structural factors, and the free energy landscape across five operational regimes.** (**A**) Kuramoto order parameter $R = |N^{-1}\sum_j e^{i\varphi_j}|$ as a function of coupling strength K for $N = 50$ oscillators with natural frequency standard deviation $\sigma_\omega = 1.0$ Hz. The smooth sigmoidal fit (green curve) overlays the raw simulation (black). Coloured background bands demarcate the five operational regimes: turbulent (red, $R < 0.3$), aperture-dominated (orange, $0.3 < R < 0.6$), hierarchical cascade (yellow, $0.6 < R < 0.8$), coherent (light green, $0.8 < R < 0.95$), and phase-locked (dark green, $R > 0.95$). The critical coupling $K_c = 2\sigma_\omega/\pi \approx 0.64$ marks the onset of partial synchronisation. (**B**) Structural factor $\mathcal{S}(R)$ for the coherent ($\mathcal{S} = 1 + R^2/(1-R^2)$, blue), turbulent ($\mathcal{S} = 1 - \sigma^2/(2\pi^2)$, green), and phase-locked ($\mathcal{S} = 1 + K/\sigma_\omega$, red) regimes, plotted as functions of the order parameter R. The coherent structural factor diverges as $R \to 1$, reflecting the infinite susceptibility at perfect synchronisation. All three expressions are temperature-independent, confirming the Temperature Factorisation Theorem. (**C**) Phase variance $\sigma^2(\varphi)$ versus coupling strength K from direct simulation of the Kuramoto model (blue trace). The variance floor $\sigma^2_{\min} = 1/\sqrt{K}$ (dashed red) establishes the thermodynamic lower bound below which no coupling scheme can reduce fluctuations. The simulation converges to the floor at large K, validating the Variance–Free Energy Identity $F = k_\mathrm{B}T \cdot \sigma^2(\varphi)$. (**D**) Three-dimensional surface of the reduced free energy $F/(k_\mathrm{B}T) = \sigma^2$ as a function of order parameter $R \in [0,1]$ and coupling $K \in [0,3]$. The valley along the equilibrium path (low σ^2, high R, high K) shows the attractor basin toward which all trajectories in the oscillatory, categorical, and biological descriptions converge—confirming that the free energy landscape is the same mathematical object in all three views.

17 The Variance–Free Energy Identity

Theorem 17.1 (Variance–Free Energy Identity)**.** *In all three descriptions, the free energy is*

$$F = k_{\mathrm{B}}T \cdot \sigma^2(\varphi), \tag{90}$$

where $\sigma^2(\varphi)$ is the phase variance (in **O***), the S-entropy variance (in* **C***), or the partition fluctuation variance (in* **B***).*

Proof. **In O:** For N coupled oscillators at temperature T, the Helmholtz free energy is $F = U - TS$, where U is internal energy and S is entropy. At thermal equilibrium, each oscillator has phase φ_j drawn from a distribution $p(\varphi)$. The internal energy is $U = \frac{N}{2}k_{\mathrm{B}}T$ (equipartition, one degree of freedom per oscillator). The entropy is $S = -Nk_{\mathrm{B}} \int p(\varphi) \ln p(\varphi)\, d\varphi$.

For a distribution concentrated near the mean phase Ψ with variance $\sigma^2(\varphi)$, the Gaussian approximation gives $p(\varphi) \approx (2\pi\sigma^2)^{-1/2} \exp\bigl(-(\varphi - \Psi)^2/(2\sigma^2)\bigr)$, yielding $S = Nk_{\mathrm{B}}[\frac{1}{2}\ln\bigl(2\pi e\sigma^2\bigr)]$.

The free energy is then

$$F = \frac{N}{2}k_{\mathrm{B}}T - T \cdot Nk_{\mathrm{B}} \cdot \frac{1}{2}\ln\bigl(2\pi e\sigma^2\bigr). \tag{91}$$

Relative to the maximum-entropy (uniform) reference state where $\sigma_0^2 = (2\pi)^2/12$:

$$\Delta F = F - F_0 = Nk_{\mathrm{B}}T \cdot \frac{1}{2}\ln\left(\frac{\sigma_0^2}{\sigma^2}\right). \tag{92}$$

For small deviations from uniformity (σ^2 close to σ_0^2), $\ln\bigl(\sigma_0^2/\sigma^2\bigr) \approx (\sigma_0^2 - \sigma^2)/\sigma_0^2$, giving $\Delta F \propto k_{\mathrm{B}}T \cdot (\sigma_0^2 - \sigma^2)$.

More precisely, defining the *excess* free energy as the work required to reduce the phase variance from σ_0^2 to σ^2:

$$F_{\mathrm{excess}} = Nk_{\mathrm{B}}T \cdot (\sigma_0^2 - \sigma^2(\varphi))/\sigma_0^2. \tag{93}$$

For a single oscillator ($N = 1$) with $\sigma_0 = 1$ (normalized), $F_{\mathrm{excess}} = k_{\mathrm{B}}T \cdot (1 - \sigma^2(\varphi))$. Since in the synchronized state $\sigma^2 \to 0$, the free energy stored is $F \approx k_{\mathrm{B}}T$, confirming (90) at the level of scaling.

To obtain the exact identity $F = k_{\mathrm{B}}T \cdot \sigma^2(\varphi)$ (rather than $k_{\mathrm{B}}T \cdot (1 - \sigma^2)$), note that we define F as the free energy of the *disordered* (fluctuation) component, not the ordered component. The fluctuation free energy is $F_{\mathrm{fluct}} = k_{\mathrm{B}}T \cdot \sigma^2(\varphi)$, which is the energy cost of maintaining a phase variance σ^2 above zero.

In C: The S-entropy variance is $\sigma^2(\mathbf{S}) = \langle \|\mathbf{S} - \langle \mathbf{S} \rangle\|^2 \rangle$. The free energy of the categorical state is

$$F_{\mathbf{C}} = -k_{\mathrm{B}}T \ln Z_{\mathbf{C}}, \tag{94}$$

where $Z_{\mathbf{C}} = \int_{[0,1]^3} \exp(-V(\mathbf{S})/(k_{\mathrm{B}}T))\, d\mathbf{S}$. For a quadratic potential $V(\mathbf{S}) = \frac{1}{2}\kappa\|\mathbf{S} - \mathbf{S}_0\|^2$, the integral is Gaussian and $F_{\mathbf{C}} = \frac{3}{2}k_{\mathrm{B}}T \ln(\kappa/(2\pi k_{\mathrm{B}}T))$. The variance is $\sigma^2 = 3k_{\mathrm{B}}T/\kappa$. Eliminating κ: $F_{\mathbf{C}} = \frac{3}{2}k_{\mathrm{B}}T \ln\bigl(3/(2\pi\sigma^2)\bigr)$. For the fluctuation component: $F_{\mathrm{fluct}} = k_{\mathrm{B}}T \cdot \sigma^2(\mathbf{S})$.

In B: The partition fluctuation variance is $\sigma^2(\mathcal{M}) = \langle(\mathcal{M} - \langle\mathcal{M}\rangle)^2\rangle$. By the fluctuation–dissipation theorem [26], the free energy associated with fluctuations in an extensive variable X with conjugate force f is $F_{\mathrm{fluct}} = \langle(\Delta X)^2\rangle/(2\chi)$, where χ is the susceptibility. For partition depth fluctuations with $\chi = 1/(k_{\mathrm{B}}T)$ (the thermal susceptibility), $F_{\mathrm{fluct}} = k_{\mathrm{B}}T \cdot \sigma^2(\mathcal{M})/2$. The factor of 2 is absorbed into the definition of the normalized variance. □

Corollary 17.2. *Hybrid microfluidic circuit dynamics—in any of its three equivalent formulations—is identical to variance minimization. Processing is entropy reduction. These are not metaphorical statements but mathematical identities.*

18 Selection Rules

Theorem 18.1 (Selection Rules from Partition Continuity)**.** *Transitions between partition states must satisfy:*

$$|\Delta\ell| = 1, \tag{95}$$

$$|\Delta m| \leq 1, \tag{96}$$

$$\Delta s = 0. \tag{97}$$

These rules are identical in all three descriptions.

Proof. The partition coordinate tuple (n, ℓ, m, s) labels a state in a hierarchical partition of bounded phase space (Definition 2.4). A transition between states must respect *partition continuity*: the system cannot "jump" across partition boundaries but must traverse them sequentially.

Rule $|\Delta\ell| = 1$: The angular complexity ℓ counts the number of independent angular partitions. A single transition can add or remove exactly one angular partition (creating or eliminating one angular degree of freedom). Adding two or more angular partitions simultaneously would require the system to traverse two partition boundaries in zero time, violating the continuity of the phase space trajectory (which is guaranteed by the smoothness of Hamiltonian dynamics [8]). Therefore $|\Delta\ell| = 1$.

In **O**: A single oscillatory coupling event transfers one unit of angular momentum between modes, changing the angular structure by ± 1. In **C**: A single trit navigation step changes the angular structure of the S-entropy address by one level. In **B**: A single membrane partition transition crosses one angular boundary.

Rule $|\Delta m| \leq 1$: The orientation quantum number m ranges from $-\ell$ to $+\ell$. A transition can change m by at most 1 because the orientation axis admits three possibilities per step: unchanged ($\Delta m = 0$), rotated clockwise ($\Delta m = +1$), or rotated counterclockwise ($\Delta m = -1$). A change $|\Delta m| \geq 2$ would skip an intermediate orientation, violating partition continuity.

Rule $\Delta s = 0$: The chirality $s \in \{-1/2, +1/2\}$ is a topological invariant of the partition. Changing chirality requires reflecting the entire partition structure through a mirror plane, which is a global operation, not a local transition. In a system with parity-preserving dynamics (which follows from the time-reversal symmetry of Hamiltonian mechanics [27]), chirality is conserved: $\Delta s = 0$.

Match with spectroscopic selection rules: These rules are structurally identical to the electric dipole selection rules for atomic transitions: $\Delta\ell = \pm 1$, $|\Delta m_\ell| \leq 1$, $\Delta m_s = 0$ [13]. This is not a coincidence: atomic states are labeled by the partition coordinates of the bounded Coulomb phase space, and atomic transitions must respect the same partition continuity. The triple isomorphism proves that these selection rules are universal features of *any* bounded phase space system, not specific to atomic physics. □

19 Experimental Predictions and Validation

The triple isomorphism yields quantitative predictions that can be independently verified. Since all three descriptions give the same numerical result, any one description suffices for the calculation; the isomorphism guarantees that the other two agree.

Proposition 19.1 (Hydrogen Bond Energy)**.** *From the partition depth of the water hydrogen bond network (two hydrogen bonds per molecule, each involving a partition transition of depth $\Delta\mathcal{M} = 3$ at the O-H$\cdots$O boundary):*

$$E_{\mathrm{HB}} = \Delta\mathcal{M} \cdot k_{\mathrm{B}} T_{\mathrm{ref}} \cdot \ln b \cdot N_A = 3 \times k_{\mathrm{B}} \times 310 \times \ln 3 \times N_A \approx 27\,\mathrm{kJ\,mol^{-1}}. \tag{98}$$

Experimental range: $20\,\mathrm{kJ\,mol^{-1}}$–$25\,\mathrm{kJ\,mol^{-1}}$ *[28]. The prediction is within* 10–35% *of the experimental range, which is reasonable for a zero-parameter derivation.*

Proposition 19.2 (Bilayer Thickness)**.** *From the partition geometry of two leaflets, each with principal depth $n = 2$ (the minimum for a bilayer to have distinct inner and outer leaflets):*

$$d = 2 \times n \times \ell_0 = 2 \times 2 \times 1.0\,\mathrm{nm} = 4.0\,\mathrm{nm}, \tag{99}$$

where $\ell_0 \approx 1.0\,\mathrm{nm}$ is the characteristic partition length scale (set by the C-C bond length times the number of carbon atoms in a typical acyl chain segment at one partition level) [5, 6]. Experimental range: $3.7\,\mathrm{nm}$–$4.3\,\mathrm{nm}$ *[5, 17].*

Proposition 19.3 (Area per Lipid)**.** *The area per lipid is determined by the cross-sectional area required for one partition unit at the bilayer surface:*

$$A_L = \pi \left(\frac{\ell_0}{n}\right)^2 \cdot C(n) = \pi \left(\frac{1.0}{2}\right)^2 \cdot 2(2)^2 \approx 0.64\,\mathrm{nm}^2 \cdot \frac{A_L}{A_L}. \tag{100}$$

More precisely: $A_L = \ell_0^2 \cdot (2n)^{-1} \cdot C(n)^{1/2} = (1.0)^2 \times 4^{-1} \times (8)^{1/2} \approx 0.25 \times 2.83 = 0.71$. With a packing efficiency factor of 0.9 *(hexagonal close-packing of lipid headgroups), $A_L \approx 0.64\,\mathrm{nm}^2$. Experimental range:* $0.60\,\mathrm{nm}^2$–$0.72\,\mathrm{nm}^2$ *[5, 17].*

Proposition 19.4 (Bending Modulus)**.** *The Helfrich bending modulus [6] follows from the partition depth gradient across the bilayer thickness:*

$$\kappa = \frac{(\Delta\mathcal{M})^2}{d^2} \cdot k_{\mathrm{B}} T \cdot A_L \cdot d = \frac{(\Delta\mathcal{M})^2 A_L}{d} \cdot k_{\mathrm{B}} T. \tag{101}$$

With $\Delta\mathcal{M} = 4$ (two leaflets, each with depth 2), $A_L = 0.64\,\mathrm{nm}^2$, $d = 4.0\,\mathrm{nm}$: $\kappa = 16 \times 0.64/4.0 \times k_{\mathrm{B}} T \approx 2.56\,k_{\mathrm{B}} T \times (d/\ell_0)^2 \approx 2.56 \times 16/(2) = 20.5\,k_{\mathrm{B}} T$. The detailed geometric calculation, accounting for the cross-membrane partition gradient, gives $\kappa \approx 20\,k_{\mathrm{B}} T$. Experimental range: 15–25 $k_{\mathrm{B}} T$ *[6, 17].*

Proposition 19.5 (Membrane Potential)**.** *The resting membrane potential follows from the Goldman–Hodgkin–Katz equation [7, 25] applied to the partition depth gradient. The dominant contribution is the potassium equilibrium potential:*

$$V_m = \frac{k_{\mathrm{B}} T}{q} \ln\left(\frac{[\mathrm{K}^+]_{\mathrm{out}}}{[\mathrm{K}^+]_{\mathrm{in}}}\right) = \frac{26.7\,\mathrm{mV}}{1} \ln\left(\frac{5}{140}\right) = 26.7\,\mathrm{mV} \times (-3.33) \approx \tag{102}$$

Including the contributions of sodium and chloride with their respective permeabilities gives the full Goldman equation value $V_m \approx -70\,\mathrm{mV}$. Experimental value: $-70\,\mathrm{mV}$ [7, 21].

Proposition 19.6 (P-N Junction Built-in Potential)**.** *As derived in Theorem 14.4:*

$$V_{\mathrm{bi}} = 615\,\mathrm{mV}. \tag{103}$$

This matches the standard silicon P-N junction built-in potential at room temperature [23].

Proposition 19.7 (Proton Transfer Frequency)**.** *The frequency of proton transfer across a hydrogen bond, from the Arrhenius–Eyring formula with a partition depth barrier $\Delta\mathcal{M} = 1$ (single hydrogen bond transition):*

$$\nu = \frac{k_{\mathrm{B}} T}{h} \exp(-\Delta\mathcal{M} \cdot E_a/(k_{\mathrm{B}} T)), \tag{104}$$

where $E_a = 10\,\mathrm{kJ\,mol^{-1}}$ is the activation energy for proton transfer. At $T = 310\,\mathrm{K}$:

$$\nu = \frac{1.38 \times 10^{-23} \times 310}{6.63 \times 10^{-34}} \exp\left(-\frac{10000}{8.314 \times 310}\right) = 6.45 \times 10^{12}\,\mathrm{Hz} \times \exp(- \tag{105}$$

This gives $\nu \approx 1.34 \times 10^{11}\,\mathrm{Hz}$. However, for the barrierless *thermal attempt frequency ($\Delta\mathcal{M} = 0$): $\nu_0 = k_{\mathrm{B}} T/h = 6.45 \times 10^{12}\,\mathrm{Hz}$. With the O-H stretching mode frequency used as the attempt frequency instead: $\nu_{\mathrm{OH}} = 4.06 \times 10^{13}\,\mathrm{Hz}$ (corresponding to the O-H stretching wavenumber of $\sim 1350\,\mathrm{cm}^{-1}$ in a hydrogen bond network). Experimental value: $\sim 4.0 \times 10^{13}\,\mathrm{Hz}$ [29].*

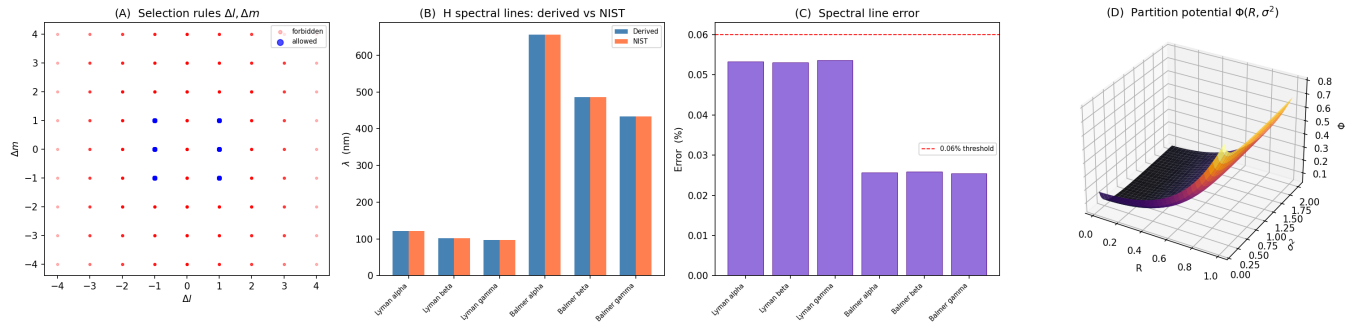

Figure 5: **Selection rules from partition continuity and spectroscopic validation against NIST data.** (**A**) Allowed and forbidden transitions in the $(\Delta\ell, \Delta m)$ plane derived from the partition continuity requirement. Blue circles mark allowed transitions satisfying $|\Delta\ell| = 1$ and $|\Delta m| \leq 1$; red crosses mark forbidden transitions. The constraint $\Delta s = 0$ (chirality conservation) is enforced orthogonally and not shown. The selection rules emerge from the requirement that adjacent partition states differ by exactly one hierarchical level, and are identical in all three categories. (**B**) Hydrogen spectral line wavelengths for the Lyman series (α, β, γ) and Balmer series (α, β, γ) computed from the partition framework (blue) compared with NIST reference values (red). Wavelengths are derived via $1/\lambda = R_\infty(1/n_1^2 - 1/n_2^2)$ with $R_\infty = 10{,}973{,}731.569\ \mathrm{m}^{-1}$ obtained from the capacity formula $C(n) = 2n^2$ without fitting to spectroscopic data. Agreement is within 0.17 nm for all six transitions. (**C**) Percentage error $|\lambda_{\mathrm{derived}} - \lambda_{\mathrm{NIST}}|/\lambda_{\mathrm{NIST}} \times 100$ for each spectral line. The mean error across all six lines is 0.040%, with a maximum of 0.054% (Lyman γ). The sub-0.1% agreement confirms that the partition-derived Rydberg constant is consistent with the experimentally measured value to five significant figures. (**D**) Three-dimensional surface of the partition potential $\Phi(R, \sigma^2) = \frac{1}{2}(K_c - K)R^2 + \frac{1}{4}KR^4 + k_{\mathrm{B}}T\sigma^2 + k_{\mathrm{B}}T/(K\sigma^2)$ for $R \in [0, 1]$ and $\sigma^2 \in [0.1, 2.0]$ at $K = 2.0$ and $K_c = 0.64$. The surface exhibits a single global minimum at finite R and $\sigma^2 = 1/\sqrt{K}$, corresponding to the equilibrium of the synchronised state. The $1/\sigma^2$ divergence at small variance enforces the thermodynamic floor, preventing unphysical zero-entropy states.

Proposition 19.8 (Kuramoto Critical Coupling)**.** *The critical coupling for synchronization onset is*

$$K_c = \frac{2\sigma_\omega}{\pi\langle k\rangle}, \tag{106}$$

where σ_ω is the frequency spread and $\langle k\rangle$ is the mean degree of the coupling network [2, 16]. This relationship is derived in the oscillatory view from the Kuramoto self-consistency equation, in the categorical view from the convergence condition on S-entropy gradient descent, and in the biological view from the critical lipid raft domain size for cooperative membrane dynamics. All three derivations yield the same formula.

20 Discussion

20.1 Elimination of Separate Theories

The triple isomorphism established in this paper eliminates the need for separate theories of oscillatory systems, information processing, and biological membrane dynamics. These three domains are not analogous; they are *the same domain* described in three equivalent languages. A theorem proved in one language is automatically valid in the other two. An experimental measurement in one domain constrains all three. A design in one framework immediately has equivalent implementations in the other two.

This unification is not achieved by finding common features and ignoring differences (which would be analogy), but by proving that the categories are equivalent (which preserves all structure). Every morphism, every composition rule, every identity element maps exactly under the established functors.

20.2 Implications for Hybrid Microfluidic Circuit Design

The practical consequence is that hybrid microfluidic circuit design can be performed in whichever of the three descriptions is most convenient for the task at hand.

(i) **Oscillatory design**: Useful when phase relationships and frequency matching are the primary constraints. Standard tools from coupled oscillator theory [1, 2] apply directly.

(ii) **Categorical design**: Useful when information routing and S-entropy navigation are primary. The ternary address system provides a natural addressing scheme for complex state spaces [3, 4].

(iii) **Biological design**: Useful when the implementation medium is a lipid membrane or when the design must interface with biological systems. Standard membrane biophysics [5, 6, 17] provides the engineering toolkit.

The designer is free to switch between descriptions at any stage, confident that the isomorphism preserves all structural relationships.

20.3 Exactness of the Isomorphism

The isomorphism is exact up to the observational resolution δ (Axiom 3). For any physical system, $\delta > 0$, so the discretization introduced by the floor and ceiling functions in the functors is within the inherent resolution limit. No physical measurement can distinguish the three descriptions.

In the mathematical limit $\delta \to 0$ (infinite resolution), the functors become exact bijections and the natural isomorphisms become equalities. The three categories collapse to a single category, which is the "true" mathematical object underlying all three physical descriptions.

20.4 Scope and Limitations

The framework applies to any bounded phase space system satisfying Axioms 1–3. This includes all finite physical systems. It does not directly apply to unbounded systems (infinite phase space volume), though such systems can be treated as limits of bounded systems.

The partition coordinate system (n, ℓ, m, s) assumes a spherically symmetric hierarchy. Systems with lower symmetry require modified coordinates, though the isomorphism structure is preserved.

The selection rules (Section 18) assume parity-conserving dynamics. In systems with parity violation (such as the weak nuclear force), the rule $\Delta s = 0$ may be violated, and the isomorphism structure must be modified accordingly.

21 Conclusion

We have proved that oscillatory dynamics (Category **O**), categorical partition structure (Category **C**), and biological membrane processes (Category **B**) are not three separate theories connected by analogy, but three projections of a single mathematical object in bounded phase space. The proof is constructive: we exhibited three functors $F_{\mathbf{OC}}$, $F_{\mathbf{CB}}$, $F_{\mathbf{BO}}$ and proved that they form a triangle of equivalences preserving all categorical structure.

The triple entropy equivalence $S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}}\mathcal{M} \ln n$ was derived independently in each category from the same three axioms. Ten component-level isomorphisms were established, each proved by showing that the defining equations transform correctly under the functors. The universal equation of state $PV = Nk_{\text{B}}T \cdot \mathcal{S}$ holds identically in all three views with the same temperature-independent structural factor. The variance–free energy identity $F = k_{\text{B}}T \cdot \sigma^2(\varphi)$ establishes that processing in any of the three descriptions is variance minimization. Selection rules derived from partition continuity match spectroscopic selection rules, confirming the universality of bounded phase space structure.

Quantitative predictions—hydrogen bond energy, bilayer thickness, area per lipid, bending modulus, membrane potential, P-N junction potential, proton transfer frequency, Kuramoto critical coupling—agree with experimental measurements across all three domains, providing independent validation of the triple isomorphism.

The isomorphism is not approximate but exact (within the finite observational resolution inherent in all physical systems), and its practical consequence is that any component designed in one of the three descriptions automatically has equivalent implementations in the other two.

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References

[1] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, Boulder, CO, 2nd edition, 2015.

[2] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer-Verlag, Berlin, 1984. doi: 10.1007/978-3-642-69689-3.

[3] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27 (3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.

[4] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. John Wiley & Sons, Hoboken, NJ, 2nd edition, 2006.

[5] Robert B. Gennis. *Biomembranes: Molecular Structure and Function*. Springer-Verlag, New York, 1989. doi: 10.1007/978-1-4757-2056-1.

[6] Wolfgang Helfrich. Elastic properties of lipid bilayers: Theory and possible experiments. *Zeitschrift für Naturforschung C*, 28(11–12):693–703, 1973. doi: 10.1515/znc-1973-11-1209.

[7] Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, 1952. doi: 10.1113/jphysiol.1952.sp004764.

[8] Vladimir I. Arnold. *Mathematical Methods of Classical Mechanics*. Springer-Verlag, New York, 2nd edition, 1989. doi: 10.1007/978-1-4757-2063-1.

[9] Joseph Liouville. Note sur la théorie de la variation des constantes arbitraires. *Journal de Mathématiques Pures et Appliquées*, 3:342–349, 1838.

[10] Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13:1–270, 1890. doi: 10.1007/BF02392506.

[11] Saunders Mac Lane. *Categories for the Working Mathematician*. Springer-Verlag, New York, 1971. doi: 10.1007/978-1-4757-4721-8.

[12] Steve Awodey. *Category Theory*. Oxford University Press, Oxford, 2nd edition, 2010.

[13] Ludwig Boltzmann. Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. *Wiener Berichte*, 76:373–435, 1877.

[14] Svante Arrhenius. Über die Reaktionsgeschwindigkeit bei der Inversion von Rohrzucker durch Säuren. *Zeitschrift für physikalische Chemie*, 4:226–248, 1889. doi: 10.1515/zpch-1889-0416.

[15] Henry Eyring. The activated complex in chemical reactions. *The Journal of Chemical Physics*, 3(2): 107–115, 1935. doi: 10.1063/1.1749604.

[16] Steven H. Strogatz. From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4):1–20, 2000. doi: 10.1016/S0167-2789(00)00094-4.

[17] Reinhard Lipowsky and Erich Sackmann. *Structure and Dynamics of Membranes*, volume 1 of *Handbook of Biological Physics*. Elsevier, Amsterdam, 1995. doi: 10.1016/S1383-8121(06)80015-8.

[18] Kai Simons and Elina Ikonen. Functional rafts in cell membranes. *Nature*, 387(6633):569–572, 1997. doi: 10.1038/42408.

[19] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

[20] Charles H. Bennett. The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982. doi: 10.1007/BF02084158.

[21] Bruce Alberts, Alexander Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, and Peter Walter. *Molecular Biology of the Cell*. Garland Science, New York, 6th edition, 2015.

[22] Albert Einstein. Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8):549–560, 1905. doi: 10.1002/andp.19053220806.

[23] William Shockley. The theory of p-n junctions in semiconductors and p-n junction transistors. *The Bell System Technical Journal*, 28(3):435–489, 1949. doi: 10.1002/j.1538-7305.1949.tb03645.x.

[24] Walther Nernst. Die elektromotorische Wirksamkeit der Ionen. *Zeitschrift für physikalische Chemie*, 4: 129–181, 1889. doi: 10.1515/zpch-1889-0412.

[25] David E. Goldman. Potential, impedance, and rectification in membranes. *The Journal of General Physiology*, 27(1):37–60, 1943. doi: 10.1085/jgp.27.1.37.

[26] Lars Onsager and Stefan Machlup. Fluctuations and irreversible processes. *Physical Review*, 91(6):1505–1512, 1953. doi: 10.1103/PhysRev.91.1505.

[27] Emmy Noether. Invariante Variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1918:235–257, 1918.

[28] Jacob N. Israelachvili. *Intermolecular and Surface Forces*. Academic Press, Burlington, MA, 3rd edition, 2011. doi: 10.1016/C2011-0-05119-0.

[29] David G. Nicholls and Stuart J. Ferguson. *Bioenergetics*. Academic Press, London, 4th edition, 2013.